

# Chapter 7

## Bank regulation and supervision

### Learning objectives

- To understand the rationale for financial regulation
- To appreciate different types of regulation
- To understand the elements of the financial safety net
- To understand the limitation and costs of regulation
- To understand the causes for regulatory reform
- To understand bank capital regulation
- To understand the increased importance of the international dimension

### 7.1 Introduction

The regulation of financial markets in general, and of banking institutions in particular, is considered a controversial issue. The financial sector is one of the most heavily regulated sectors in the economy and banking is by far the most heavily regulated industry. In Chapter 1, we presented some of the reasons why banks are considered 'special', outlined the existence of market imperfections (such as information asymmetries, moral hazard and adverse selection) and noted how the existence of banks can help minimise such problems. In this chapter we note the pivotal role played by banks in the economy to understand the rationale for regulation (Section 7.2). We will discuss the aims and objectives and the different types of regulation in Section 7.3. Section 7.4 will discuss the elements of the financial safety net. The limitations of regulation and the possible reasons behind regulatory failure are discussed in Section 7.5. Section 7.6 reviews the causes of regulatory reform. International policy initiatives, such as the Basel Capital Adequacy Accords, are reviewed in Sections 7.7 to 7.12.

Before we discuss the rationale for regulation it is useful to introduce various terms that are often used to describe the regulatory environment. **Regulation** relates to the setting of specific rules of behaviour that firms have to abide by – these may be set through legislation (laws) or be stipulated by the relevant regulatory agency. **Monitoring** of these regulations refers to the process whereby the relevant authority assesses financial firms to evaluate whether these rules are being obeyed. **Supervision** is a broader term used to refer to the

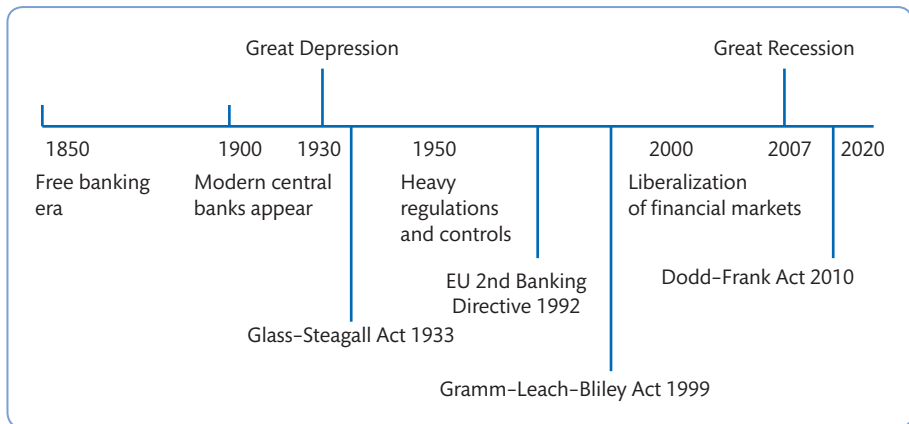
general oversight of the behaviour of financial firms. In practice, one should note that these terms are often used interchangeably in general discussion of the regulatory environment.

## 7.2 The rationale for regulation

There is a vast body of financial regulation in existence. This is normally extended incrementally, frequently to close a loophole that some earlier fraud has exposed. Financial regulation is often (but not always) enacted in response to a crisis. For example, the Glass–Steagall Act (1933) introduced the separation of bank types according to their business (commercial and investment banking) and founded the Federal Deposit Insurance Corporation (FDIC) for insuring bank deposits in response to widespread bank failures and massive losses to bank customers. Figure 7.1 illustrates a broad timeline of some of the key milestones in financial regulation over the last two centuries.

Financial systems are prone to periods of instability. A number of financial crises around the world (South East Asia, Latin America and Russia) have brought about a large number of bank failures. Further, the 2007–2009 crisis and the widespread financial turmoil that it caused brought these issues to the fore once again. These occurrences, some argue, suggest a case for more effective regulation and supervision. Others attribute many of these crises, including the global financial crisis of 2007–2009, to the failure of regulation. Advocates of the so-called ‘free banking’ argue that the financial sector would work better without regulation, supervision and central banking.<sup>1</sup>

In the absence of government regulation, they argue, banks would have greater incentives to prevent failures. However, the financial services industry is a politically sensitive one and largely relies on public confidence. Because of the nature of their activities (illiquid assets and short-term liabilities), banks are more prone to troubles than other firms. Further, because of the interconnectedness of banks, the failure of one institution can immediately affect others.



**Figure 7.1** A timeline of financial regulation

<sup>1</sup> See Chapter 6 for more discussion on free banking.

This is known as bank **contagion** and may lead to **bank runs**. Banking systems are vulnerable to **systemic risk**, which is the risk that problems in one bank will spread through the whole sector (bank failures and financial crises will be analysed in detail in Chapter 8).

Bank runs occur when a large number of depositors, fearing that their bank is unsound and about to fail, try to withdraw their savings within a short period of time. A bank run starts when the public begins to suspect that a bank may become insolvent. This creates a problem because banks keep only a small fraction of deposits in cash – they lend out the majority of deposits to borrowers or use the funds to purchase other interest-bearing assets. When a bank is faced with a sudden increase in withdrawals, it needs to increase its liquidity to meet depositors' demands. Banks' reserves may not be sufficient to cover the withdrawals and banks may be forced to sell their assets. Banks' assets (loans) are highly illiquid in the absence of a secondary market and if banks have financial difficulties they may be forced to sell loans at a loss (known as 'fire-sale' prices in the United States) in order to obtain liquidity. However, excessive losses made on such loan sales can make the bank insolvent and bring about bank failure.

Bank loans are highly illiquid because of information asymmetries: it is very difficult for a potential buyer to evaluate customer-specific information on the basis of which the loan was agreed. The very nature of banks' contracts can turn an illiquidity problem (lack of short-term cash) into insolvency (where a bank is unable to meet its obligations – or to put this differently – when the value of its assets is less than its liabilities).

In summary, regulation is needed to ensure consumers' confidence in the financial sector. The main reasons for financial sector regulation can be summarised as follows:

- to ensure systemic stability;
- to ensure the safety and soundness of individual financial institutions;
- to provide smaller, retail clients with protection, and to protect consumers against monopolistic exploitation.

**Systemic stability** is one of the main reasons for regulation, as the social costs of bank failure are greater than the private costs. Financial stability has become a common term since the 2007–2009 financial crisis. While definitions might differ, at the core, financial stability implies that the financial system – which includes financial intermediaries, markets and market infrastructures – is capable of withstanding shocks, whatever the origin of such shocks, without major disruption to the real economy. The second concern is to maintain the safety and soundness of individual financial institutions. This concern is at the core of **micro-prudential regulation**, which relies on the view that if banks do not take excessive risks, the financial system will be safe and the real economy will not suffer from a financial crisis. However, this approach failed during the global financial crisis because it ignored systemic risk. The final concern relates to **consumer protection**. In financial markets *caveat emptor* (a Latin phrase meaning 'Let the buyer beware') is not considered adequate, as financial contracts are often complex and opaque. The costs of acquiring information are high, particularly for small, retail customers. Consumer protection is a particularly sensitive issue if customers face the loss of their lifetime savings. Finally, regulation serves the purpose of protecting consumers against the abuse of monopoly power in product pricing. Table 7.1 illustrates the objectives, the reasons, the economic rationale and the costs of financial regulation. The case for regulation depends on various market imperfections and failures, which, in absence of regulation, produce sub-optimal results and reduce consumer welfare. The purpose of regulation should be limited to correcting for identified market imperfections and failures.

**Table 7.1** Bank regulation: key concepts

| Objectives                                                  | Reasons                                       | Rationale                         | Costs                                   |
|-------------------------------------------------------------|-----------------------------------------------|-----------------------------------|-----------------------------------------|
| Sustain systemic stability                                  | Key position of banks in the financial system | Market failures and imperfections | Moral hazard                            |
| Maintain the safety and soundness of financial institutions | Consumer demand                               | Potential systemic problems       | Agency capture                          |
| Protect consumer                                            |                                               | Monitoring of financial firms     | Compliance costs<br>Costs of entry/exit |
|                                                             |                                               | Ensuring consumers' confidence    | Control over products/activities/prices |

## 7.3 Types of regulation

It is possible to identify three different types of regulation:

- systemic (or macro-prudential) regulation;
- prudential (or micro-prudential) regulation;
- conduct of business regulation.

### 7.3.1 Systemic (macro-prudential) regulation

In response to the lessons of the global financial crisis, **macro-prudential regulation** is now a key financial policy area. Macro-prudential regulation aims to mitigate risk to the financial system as a whole and uses primarily prudential tools to limit systemic risk and to mitigate its effects on the real economy.

The macro-prudential approach is 'top-down': common exposures among institutions and correlations are important. More specifically, macro-prudential policies have two main objectives:

- to strengthen the resilience of the financial system; and
- to limit the build-up of systemic financial risks.

Macro-prudential policies focus on the interaction between the financial institutions, markets, infrastructure and the wider economy. In addition, macroprudential policies aim to ensure that large, systemically important financial institutions (SIFIs) are subject to more stringent prudential requirements and supervision than smaller players. This is because the potential failure of such institutions would have more serious consequences to the financial system and the economy as a whole (see also Chapter 8). Prudential regulations target individual institutions, regardless of the institution's impact on the financial system as a whole, and tend to apply uniformly across all institutions in the financial system, regardless of the prevailing macroeconomic conditions at any given point in time: the micro-prudential approach assumes that the sources of risk are exogenous and independent of the collective impact of the interactions between individual institutions. Importantly, prudential guidelines apply to regulated entities (i.e. banks) and do not often cover other financial institutions.

However, macro-prudential policies aim to address the system as a whole. In this context, a macro-prudential policy framework would simultaneously consider both the cross-sectional dimension (i.e. the concentration of risks in the financial system) as well as the time-dimension aspects (i.e. the build-up of risk over time).

Macro-prudential regulation has three key dimensions:

- *Time dimension*: relates to policies designed to prevent the excessive build-up of risk, resulting from external factors and market failures, to smooth the financial cycle.
- *Cross-sectional dimension*: relates to policies designed to make the financial sector more resilient and limit contagion effects.
- *Structural dimension*: relates to policies designed to encourage a system-wide perspective in financial regulation to create the right set of incentives for market participants.

Macro-prudential supervision is concerned with the aggregate effect of individual banks' actions. Because it aims to generate an overall picture of the functioning of the financial sector, macro-prudential supervision is also referred to as 'top-down supervision'.

Under the heading of macro-prudential regulation, we also consider all public policy regulation designed to minimise the risk of bank runs that goes under the name of the **financial safety net**. In particular, the safety net encompasses two main features – **deposit insurance** arrangements and the **lender of last resort** (LOLR) function:

- Deposit insurance is a guarantee that all or part of the amount deposited by savers in a bank will be paid in the event that a bank fails.
- The LOLR function is one of the main functions of a central bank. The central bank, or other central institution, will provide funds to banks that are in financial difficulty and are not able to access any other credit channel. A more detailed discussion of the functions of central banks was presented in Chapter 6. Through the LOLR mechanism, the authorities can provide liquidity to the banking sector at times of crises.

### 7.3.2 Prudential regulation

Prudential regulation is mainly concerned with consumer protection. It relates to the monitoring and supervision of financial institutions, with particular attention paid to asset quality and **capital adequacy**. The case for prudential regulation is that consumers are not in a position to judge the safety and soundness of financial institutions due to imperfect consumer information and agency problems associated with the nature of the intermediation business.

- *Micro-prudential regulation* – the objective of micro-prudential regulation is to limit the distress of individual financial institutions. The micro-prudential regulatory approach is 'bottom up' – in this context correlations and common exposure across institutions are of secondary importance. Micro-prudential supervision checks that individual financial firms are complying with financial regulation and involves the collection and analysis of information about the risks that the firms take, their systems and their personnel. Because micro-prudential supervision uses firm-specific information to generate a picture of risk and its management, it is also referred to as 'bottom-up supervision'.

- *Conduct of business regulation* – this focuses on how banks and other financial institutions conduct their business. This kind of regulation relates to information disclosure, fair business practices, competence, honesty and integrity of financial institutions and their employees. Overall, it focuses on establishing rules and guidelines to reduce the likelihood that:
  - consumers receive bad advice (possible agency problem);
  - supplying institutions become insolvent before contracts mature;
  - contracts turn out to be different from what the customer was anticipating;
  - fraud and misrepresentation take place;
  - employees of financial intermediaries and financial advisors act incompetently;
  - insider trading takes place;
  - money will be laundered.

### 7.3.3 Micro- and macro-prudential policies

Micro- and macro-prudential policies usually complement each other in pursuit of their respective goals. For example, the safety and soundness of individual institutions is a necessary, though not sufficient, condition for systemic stability. Similarly, a stable system contributes to the health of individual institutions. However, conflict might occur because of overlapping policy mandates and the way in which policies are applied. Both micro-prudential and macro-prudential authorities might use the same policy instruments, but they have different objectives. For example, both use buffers (capital and liquidity), but micro-prudential policy adjusts capital based on individual institutions' risks, while macro-prudential policy adjusts overall levels of capital based on the financial cycle and systemic relevance to guard against systemic risk build-up (Osiński *et al.*, 2013). Table 7.2 summarises the main policy toolkits.

**Table 7.2** Micro- and macro-prudential policy tools

| Instrument                     | Micro-prudential | Macro-prudential |
|--------------------------------|------------------|------------------|
| Capital risk weights           | Yes              | Yes              |
| Pillar 2 capital requirements  | Yes              | Yes              |
| Countercyclical capital buffer | No               | Yes              |
| Capital conservation buffer    | Yes              | No               |
| Systemic capital surcharge     | No               | Yes              |
| Dynamic provisioning           | Yes              | Yes              |
| Leverage ratio                 | Yes              | Yes              |
| Large exposure limits          | Yes              | Yes              |
| Loan-to-value limits           | Yes              | Yes              |
| Debt-to-income limits          | Yes              | Yes              |
| Foreign exchange limits        | Yes              | Yes              |
| Liquidity requirements         | Yes              | Yes              |
| Risk management standards      | Yes              | No               |
| Licensing standards            | Yes              | No               |

## 7.4 The financial safety net

One of the consequences of the 2007–2009 global financial crisis is that more attention is now being paid to financial regulation, and that holds true both regarding the protection of the stability of individual banks (micro-prudential regulation) as well as the system overall (macro-prudential regulation). A financial safety net is a comprehensive system for enhancing and ensuring a country's financial stability. It often consists of five elements (see Figure 7.2), which complement and strengthen each other (Bernet and Walter, 2009):

- regulation and supervision;
- deposit insurance schemes;
- lender of last resort;
- bank insolvency/resolution laws;
- co-operation and resolution processes.

### 7.4.1 Deposit insurance

As mentioned, deposit insurance is a guarantee that all or part of the amount deposited by savers in a bank will be paid in the event that the bank fails. The guarantee may be either explicitly given in law or regulation, offered privately without government backing or may be inferred implicitly from the verbal promises and/or past actions of the authorities. The level and coverage of deposit insurance is country specific. In some countries there are no statutory rules regarding the eligibility of bank liabilities, the level of protection provided or the form which reimbursement will take. The United States was the second country (after



**Figure 7.2** Elements of the financial safety net

Source: Adapted from Bernet and Walter (2009).

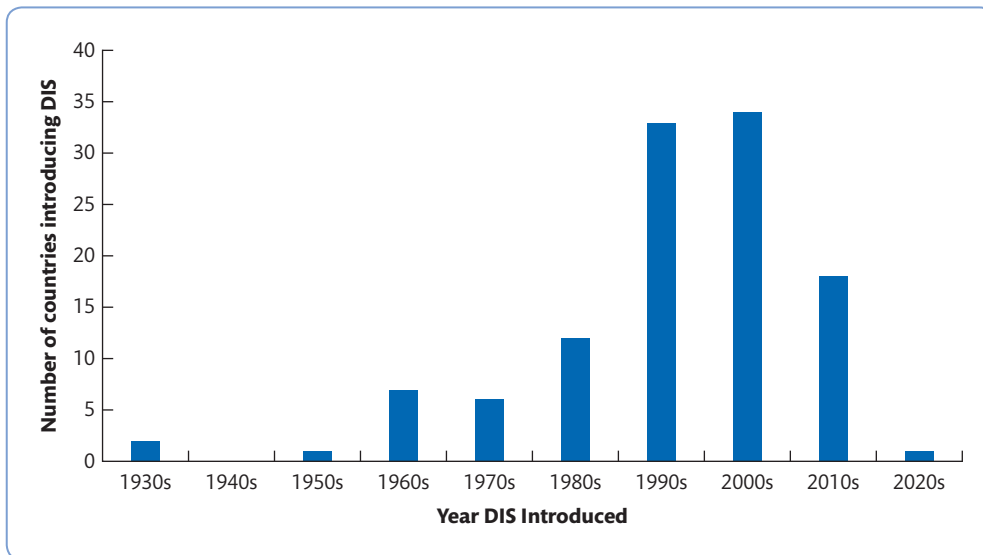


Czechoslovakia) to introduce deposit insurance in the 1930s; a substantial number of countries adopted deposit insurance schemes/systems (DIS) between 1970 and 2011.<sup>2</sup>

According to the International Association of Deposit Insurers (IADI), 146 countries had deposit insurance schemes in place in 2020, up from 111 in 2011. In Europe, the first EU Directive on deposit guarantee schemes was issued in 1994 (Directive 94/19/EC), setting minimum levels of deposit insurance that had to be in place. A number of countries did not have an explicit DIS in the run up to the 2007–2009 global financial crisis: for example, Australia, China, Saudi Arabia and South Africa, among others, had no explicit deposit protection systems.

Figure 7.3 illustrates a timeline of the introduction of DIS across the world – we can see that most countries set up their own schemes in the 1990s and 2000s, although several did so only after the financial crisis.

The crisis also exposed some fundamental weaknesses in the deposit insurance schemes operating at the time. In addition, for a number of years the IMF and the World Bank have been posing questions concerning the adjustment of deposit insurance schemes to the rapidly changing national and international conditions of financial markets. In nearly all countries hit by the crisis, there began a fundamental revision of the existing deposit insurance schemes. The financial crisis also caused a range of other supranational institutions to concern themselves with questions relating to the structure and implementation of national deposit insurance schemes. A report by the Financial Stability Forum (FSF) in early 2008 stressed the necessity of internationally recognised basic principles and conditions for the structure of deposit insurance schemes (FSF, 2008). In response, in the summer of 2008, the Basel Committee on Banking Supervision (BCBS), together with the IADI, issued the revised version of the so-called ‘Core principles for effective deposit insurance systems’. The



**Figure 7.3** A timeline of DIS introduction

Source: Authors' elaborations. Data from The World Bank. Deposit Insurance Dataset. Available at <https://datacatalog.worldbank.org/dataset/deposit-insurance-dataset>

<sup>2</sup>The Federal Deposit Insurance Corporation (FDIC) was created in 1933 in response to the thousands of bank failures that occurred in the 1920s and early 1930s.



results of this work – a set of 18 principles in total, which together define the target function of a modern deposit insurance scheme – were presented in the summer of 2009. Since then, IADI has been collaborating with the BCBS, the European Forum of Deposit Insurers (EFDI), the IMF, the World Bank and the European Commission (EC) to develop a robust methodology to assess compliance with the Core Principles. A final version of the Core Principles was submitted to the Financial Stability Board in December 2010 (Basel Committee on Banking Supervision and International Association of Deposit Insurers, 2010).<sup>3</sup> These Core Principles are intended as a voluntary framework for effective deposit insurance practices. National authorities are free to put in place supplementary measures that they deem necessary to achieve effective deposit insurance in their jurisdictions. The Core Principles (summarised in Box 7.1) are not designed to cover all the needs and circumstances of every deposit insurance system or prescribe a single specific form of deposit insurance but are designed to be adaptable to a broad range of country circumstances, settings and structures. Following the development of the Core Principles and their assessment methodology, the FSB agreed to undertake a peer review of deposit insurance systems in 2011. While the original Core Principles served their purpose well in terms of helping jurisdictions assess their deposit insurance systems and identify areas for improvement, efforts were made to improve compliance, maintain continuity and increase. As a result, a revised set of Core Principles was published in 2014.

### BOX 7.1 CORE PRINCIPLES FOR EFFECTIVE DEPOSIT INSURANCE SYSTEMS

- 1 Public policy objectives
- 2 Mandate and powers
- 3 Governance
- 4 Relationship with other safety net participants
- 5 Cross-border issues
- 6 Deposit insurer's role in contingency planning and crisis management
- 7 Membership
- 8 Coverage
- 9 Sources and uses of funds

<sup>3</sup>The Financial Stability Board (FSB) was established in April 2009 as the successor to the Financial Stability Forum (FSF). The FSF was founded in 1999 by the G7 finance ministers and central bank governors. In November 2008, the leaders of the G20 countries called for a larger membership of the FSF. In April 2009 an expanded FSF was re-established as the FSB with a broadened mandate to promote financial stability. The FSB member institutions are the central banks and/or Treasuries (or equivalent government departments) from the following countries: Argentina, Australia, Brazil, Canada, China, France, Germany, Hong Kong SAR, India, Indonesia, Italy, Japan, Mexico, the Netherlands, Republic of Korea, Russia, Saudi Arabia, Singapore, South Africa, Spain, Switzerland, Turkey, United Kingdom and United States. It also includes the following international organisations: Bank for International Settlements, European Central Bank, International Monetary Fund, Organisation for Economic Co-operation and Development and the World Bank. In addition, the following international standard-setting bodies are also members: Basel Committee on Banking Supervision, Committee on the Global Financial System, Committee and Payment and Settlement Systems, International Association of Insurance Supervisors, International Accounting Standards Board and the International Organization of Securities Commissions.

**Box 7.1 Core principles for effective deposit insurance systems (continued)**

- 10** Public awareness
- 11** Legal protection
- 12** Dealing with parties at fault in a bank failure
- 13** Early detection and timely intervention and resolution
- 14** Effective resolution processes
- 15** Reimbursing depositors
- 16** Recoveries

Source: [www.iadi.org/en/core-principles-and-research/core-principles/](http://www.iadi.org/en/core-principles-and-research/core-principles/)

An explicit DIS is considered preferable to implicit protection if it clarifies the authorities' obligations to depositors and limits the scope for discretionary decisions that may result in arbitrary actions. A DIS should cover all those deposits that can lead to a liquidity problem for a bank; these include all kind of saving accounts and short-term deposits held by private individuals and business customers (excluding financial institutions). In the European Union, deposits are defined in Article 1(1) of Directive 94/19/EC. In 2008, the need to restore confidence in the financial sector was vital and the European Commission put forward a revision of EU rules to promote convergence of deposit guarantee schemes within member states in order to improve depositor protection, including increased level of coverage for deposits (from €20,000 to €100,000), the removal of co-insurance (i.e. where the depositor bears part of the losses) and the reduction of the payout period from three months to three days. These proposals were adopted by the European Council in February 2009. However, the evolution of the eurozone crisis highlighted the need for more drastic improvements, leading to the creation of a Single Deposit Guarantee Scheme (SDGS). The EU SDGS deal is discussed in Section 14.4.5.

In general terms, the deposits that are repayable by the DIS are called eligible deposits. Not all eligible deposits are repayable, as some countries apply a coverage limit. For example, in the UK if a bank, building society or credit union were to default, the Financial Services Compensation Scheme (FSCS) would automatically refund savings up to £85,000 (up to £170,000 for joint accounts). According to the FSCS, around 98 per cent of the UK population have less than £85,000 in savings and are therefore covered by the protection limit. The level of deposit protection in the EU is harmonised at €100,000 (or equivalent amount in the local currency).

A lesson from the 2007–2009 financial crisis is that a low level of coverage can increase financial instability. While a high level of coverage might reduce the incentive of depositors to run, it also reduces market discipline. The Core Principles do not prescribe a preferred coverage level. However, they suggest that limits should be set so that the vast majority of small-scale retail depositors are covered in full (so they have no incentive to run) but that a significant portion of the value of total deposit liabilities remains uncovered and exposed to market discipline.

To contribute effectively to the stability of a country's financial system a DIS needs to be: (i) credible; (ii) properly designed; (iii) well implemented and understood by the public; (iv) supported by strong prudential regulation and supervision; (v) supported by sound accounting and disclosure regimes; and (vi) supported by the enforcement of effective laws (Bernet and Walter, 2009; Adema *et al.*, 2019).

It is necessary to point out that a deposit insurance system can deal with a limited number of simultaneous bank failures, but cannot be expected to deal with a systemic banking crisis. The practical organisation of a DIS is country-specific; it can be either private or public, depending on a country's legal circumstances.

#### 7.4.1.1 Types of deposit insurance schemes

There are different types of deposit insurance schemes, of which the most common are the following:<sup>4</sup>

- the paybox model;
- the cost reducer model (or loss minimiser);
- the resolution facilitator model;
- the supervisor model (or risk minimiser).

In the *paybox model* the role of the deposit insurance institution is narrowly based and limited to a settlement function (i.e. limited to paying out the depositors and ensuring an orderly settlement of all claims). A number of countries adopt this narrow mandate, including Australia, Germany, Hong Kong, India, the Netherlands, Singapore and Switzerland. The UK (together with Argentina and Brazil) adopts a so-called 'paybox plus' model, where the deposit insurer has some additional responsibilities, such as a resolution function.

In the *cost reducer model*, the deposit insurance institution retains the settlement function of the paybox model and, in addition to it, takes on the role of handling any occurrence of insolvency in an insured institution with the lowest possible costs and externalities for the financial system. Nine countries adopt this loss-minimiser approach, where the deposit insurer is tasked with pursuing the least-cost resolution strategy: Canada, France, Indonesia, Italy, Japan, Mexico, Russia, Spain and Turkey.

In the *resolution facilitator model*, the deposit insurance institution has additional powers: it can intervene to support a bank in difficulties (but not an illiquid or insolvent bank); it can facilitate a corporate restructuring or even a merger in order to protect depositors.

Finally, in the *supervisor model*, the deposit insurance institution has a broad mandate, as it is part of the supervisory system (for example, the Federal Deposit Insurance Corporation (FDIC) in the US and the Korean Deposit Insurance Corporation, KDIC).

Following the 2007–2009 financial crisis, the mandates of many deposit insurers have been revised and expanded and many countries have adopted versions of the 'cost minimiser' approach. The financial crisis also illustrated that depositors' confidence depended, in part, on knowing that adequate funds would always be available to ensure the prompt reimbursement of their claims (Core Principle 11). While the primary responsibility for paying the cost of deposit insurance should be borne by banks, adequate emergency funding arrangements were also considered important (FSB, 2012d). The main issues policy makers have to address in relation to the funding of a DIS is whether it should be funded *ex ante* (where a fund is set up so that it can be drawn upon to secure payments) or *ex post* (where there is no fund and the participants in the DIS will be required to contribute in the event of a claim). Both *ex-ante* and *ex-post* funding have a number of advantages and disadvantages. *Ex post* funding fosters

<sup>4</sup>For a detailed discussion of different types of deposit insurance, see Bernet and Walter (2009) and Adema *et al.* (2019).

market discipline, as banks have an incentive to monitor each others' activities. However, there may be delays when the funds are needed. In addition, requests for contributions to the DIS may come at a time of economic instability, triggering a domino effect of bank failures.

*Ex-ante* financing, meanwhile, might boost public confidence, smooth premium payments and reduce moral hazard if it incorporates risk adjusted premiums (see Box 7.2 for a discussion of how safety net arrangements can increase moral hazard). Drawbacks of setting up an insurance fund *ex ante* include the difficulty in establishing and managing a fund of adequate size to guarantee deposits of large financial institutions. In addition, while depositors seem to have a clear preference for a DIS based on *ex-ante* financing, this type of funding requires the definition of a deposit insurance premium. The key unanswered questions in this context are how to price the deposit insurance premiums; how to arrive at a 'fair' premium; what the premium should cover and how to incorporate systemic risk in the deposit insurance premiums.

The investigation of risk-based models for computing contributions of DIS members has become a crucial goal for regulators in the aftermath of the global financial crisis. For example, the European Commission Joint Research Centre, in co-operation with the European Forum of Deposit Insurers (EFDI), has investigated potential models and assessed their potential impact across EU member states (European Commission, 2009). In addition, the Financial Stability Board (2012c) has undertaken a comprehensive review of the reforms carried out to ensure depositors protection as well as the structure of possible arrangements going forward. Table 7.3 presents a cross-country comparison of deposit insurance systems

**Table 7.3** DIS cross-country comparison

| Country     | Coverage level (US\$) <sup>1</sup> | Public policy objectives formalised    | Admin   | Multiple agencies/systems | Type/mandate   | Type/funding   |
|-------------|------------------------------------|----------------------------------------|---------|---------------------------|----------------|----------------|
| Argentina   | 7,545                              | In law                                 | Private | No                        | Paybox Plus    | <i>Ex ante</i> |
| Australia   | 1,016,300                          | In annex to law                        | Public  | No                        | Paybox         | <i>Ex ante</i> |
| Brazil      | 42,000                             | In statutes                            | Private | Yes                       | Paybox plus    | <i>Ex ante</i> |
| Canada      | 100,000                            | In law                                 | Public  | Yes                       | Loss minimiser | <i>Ex ante</i> |
| France      | 136,920                            | In law and regulation                  | Mixed   | No                        | Loss minimiser | <i>Ex ante</i> |
| Germany     | 136,920                            | In law                                 | Mixed   | Yes                       | Paybox         | <i>Ex ante</i> |
| Hong Kong   | 64,000                             | In law                                 | Public  | No                        | Paybox         | <i>Ex ante</i> |
| India       | 2,240                              | Preamble of law, annual report         | Public  | No                        | Paybox         | <i>Ex ante</i> |
| Indonesia   | 235,294                            | In law                                 | Public  | No                        | Loss minimiser | <i>Ex ante</i> |
| Italy       | 136,920                            | In law                                 | Private | Yes                       | Loss minimiser | <i>Ex ante</i> |
| Japan       | 122,775                            | In law                                 | Mixed   | Yes                       | Loss minimiser | <i>Ex ante</i> |
| Korea       | 43,902                             | In law                                 | Public  | No                        | Risk minimiser | <i>Ex ante</i> |
| Mexico      | 146,606                            | In law                                 | Public  | No                        | Loss minimiser | <i>Ex ante</i> |
| Netherlands | 136,920                            | In law                                 | Public  | No                        | Paybox         | <i>Ex post</i> |
| Russia      | 23,064                             | In law                                 | Public  | No                        | Loss minimiser | <i>Ex ante</i> |
| Singapore   | 38,835                             | Ministerial statement, preamble of law | Public  | No                        | Paybox         | <i>Ex ante</i> |
| Spain       | 136,920                            | In law                                 | Private | No                        | Loss minimiser | <i>Ex ante</i> |

Table 7.3 continued

| Country     | Coverage level (US\$) <sup>1</sup> | Public policy objectives formalised | Admin   | Multiple agencies/ systems | Type/ mandate  | Type/ funding  |
|-------------|------------------------------------|-------------------------------------|---------|----------------------------|----------------|----------------|
| Switzerland | 96,830                             | In law and statutes                 | Private | No                         | Paybox         | <i>Ex post</i> |
| Turkey      | 32,341                             | In law                              | Public  | No                         | Loss minimiser | <i>Ex ante</i> |
| UK          | 133,068                            | In law and statutes                 | Public  | No                         | Paybox plus    | <i>Ex post</i> |
| US          | 250,000                            | In law                              | Public  | Yes                        | Risk minimiser | <i>Ex ante</i> |

Notes: <sup>1</sup>Using the exchange rate as of end-2010.

Source: The information shown in this table is collated from different tables in Annex C of the Thematic Review on Deposit Insurance Systems, Financial Stability Board (2012d).

features. Overall, explicit deposit insurance is widely adopted across the world and is now a standard policy of bank regulators and supervisors. The 2007–2009 global financial crisis has reinforced the importance of deposit insurance in promoting stability and its role in the overall financial safety net provided to the banking sector (Anginer *et al.*, 2019).

### 7.4.2 Lender of last resort

The lender of last resort function is typically one of the main functions of a central bank (see Section 6.5.1). The central bank, or other central institution, will provide funds to banks that are in financial difficulty and are not able to access any other credit channel. Through the LOLR mechanism, the authorities can provide liquidity to the banking sector at times of crises. The lender of last resort normally operates at the domestic level. The growing internationalisation of banking institutions and the banking and financial crises have raised the issue of an international lender of last resort. However, there is little agreement on the desirability of such cross-country deposit insurance institution. In the case of a global scheme it is not clear whether it would be appropriate for the IMF to play such a role. However, a number of elements of a global financial safety net are at various stages of approval (the IMF, for example, has introduced a new precautionary line of credit and improved its existing flexible line of credit for countries that meet a rigorous set of criteria). The IMF publishes a *Global Financial Stability Report*, which examines current risks facing the global financial system and highlights policy actions that may mitigate these (see also Chapter 8).

### 7.4.3 Bank insolvency and resolution laws

As a part of the regulation specific to the financial industry, bank insolvency and resolution laws refer to the legal provisions that regulate the conditions for the handling of bank failures and insolvencies. Bankruptcy laws applicable for general types of companies are generally not suited for balancing or preventing the negative externalities to the real economy triggered by bank insolvency. For this reason, most countries have enforced a bankruptcy law or insolvency regime aimed specifically at banks and financial institutions (Bernet and Walter, 2009). These are also known as special resolution regimes (SRR). The design and implementation of bank insolvency regulations is subject to strong political influence and there is a wide range of approaches adopted in EU countries and elsewhere. Since the onset of the global financial crisis in 2007, many countries have enacted major regulatory reforms to develop new, or

revise existing resolution regimes. In the UK, the Banking Act of 2009 created an SRR that gives the UK authorities (HM Treasury, Bank of England and Prudential Regulation Authority) a permanent framework, providing them with tools for dealing with failing UK banks, building societies and certain investment firms.<sup>5</sup> The Act also gave the Bank of England a key role in implementing a resolution using statutory resolution tools. The PRA, in consultation with the Bank of England and the Treasury, makes the decision to put a bank into the SRR. A number of regulatory authorities then play a role in the SRR. HM Treasury decides whether to put a bank into temporary public ownership. If a bank is not temporarily nationalised, the Bank of England, in consultation with the other authorities, decides which of the tools to use and implements the resolution. This work is done by the Special Resolution Unit (SRU) within the Bank of England. The SRU was established in February 2009 to specifically deal with distressed banks and building societies. The Financial Services Compensation Scheme (FSCS) also has a role in the SRR, paying out depositors covered by its depositor compensation scheme and its funds may be used also to support a non-payout resolution provided that is no more costly to the FSCS, net of recoveries, than a payout.

The 2009 UK Banking Act sets out five key objectives that must be considered in choosing which resolution tools to use (Bank of England, 2017):

- protect and enhance financial stability;
- protect and enhance public confidence in the financial system's stability;
- protect public funds;
- protect depositors and investors covered by the FSCS;
- protect (where relevant) client assets;
- avoid interfering in property rights.

In the United States and Canada, the Federal Deposit Insurance Corporation (FDIC) and the Canada Deposit Insurance Corporation (CDIC) are subject to least-cost resolution (LCR) requirements, whereby they must adopt the resolution method which costs the least to the deposit insurance fund regardless of other objectives (although these requirements can be overridden in circumstances that trigger 'financial stability exceptions'). Some countries specify public interest objectives in their SRRs, including New Zealand and Hong Kong. Other countries, for example EU countries, do not specify financial stability of public interest objectives in the legislation relating to the resolution of failing banks.

In the EU, a Single Resolution Mechanism (SRM) was established in 2013 in the context of the creation of the Banking Union (see Chapter 14). The SRM regulation builds on the 'Rulebook on bank resolution' set out in the 2013 **Bank Recovery and Resolution Directive (BRRD)**; it comprises establishing a Single Resolution Board (SRB) and a Single Bank Resolution Fund (SRF); these are discussed in Section 14.4.4. The harmonisation of the EU legislation in this area should ensure that future bank failures could be managed with minimal disruption for financial stability and public finances. Given the present diversity in European resolution legislation and practice (see Table 7.4) this seems a necessary step forward for the EU banking systems. Even outside the EU, the growing internationalisation of banking requires a closer co-operation of national deposit insurance institutions, regulators and resolution authorities.

Table 7.4 illustrates the administrative authority responsible for restructuring banks in a number of countries. There are substantial cross-country differences, in terms of scope,

<sup>5</sup> In 2014, the resolution regime was extended to include central counterparties (CCPs).

mandates and powers of authorities, which reflect differences in the regulatory environment. While there is no preferred resolution regime, there are significant divergences and inconsistencies that make the resolution of internationally active banks extremely difficult. To address some of these issues, the Basel-based Financial Stability Board (2011a), in consultation with relevant standard-setting bodies, published 'Key attributes of effective resolution regimes for financial institutions' as part of the package of policy measures to address the moral hazard risks posed by systemically important financial institutions (SIFIs).

**Table 7.4** Bank resolution authorities: a cross-country comparison

| Country   | Resolution authority                           |                                       |                                                |                                                          |
|-----------|------------------------------------------------|---------------------------------------|------------------------------------------------|----------------------------------------------------------|
|           | Central bank and banking supervisor            | Banking supervisor                    | Deposit insurance and bank resolution agency   | Integrated financial regulator                           |
| Argentina | Banco Central de la Republica Argentina (BCRA) |                                       |                                                | Australian Prudential Regulation Authority (APRA)        |
| Australia |                                                |                                       |                                                |                                                          |
| Brazil    |                                                |                                       |                                                |                                                          |
| Canada    | Banco Central do Brasil (BCB)                  |                                       | Canada Deposit Insurance Corporation (CDIC)    |                                                          |
| China     |                                                |                                       |                                                |                                                          |
| France    |                                                |                                       |                                                |                                                          |
| Germany   |                                                | Autorité de contrôle prudentiel (ACP) |                                                | Federal Financial Supervisory Authority                  |
| Hong Kong |                                                |                                       |                                                |                                                          |
| India     |                                                |                                       |                                                |                                                          |
| Indonesia | Reserve Bank of India (RBI)                    |                                       | Indonesia Deposit Insurance Corporation (IDIC) | Federal Agency for Financial Market Stabilisation (FMSA) |
| Italy     |                                                |                                       |                                                |                                                          |
| Japan     |                                                |                                       |                                                |                                                          |
|           |                                                |                                       |                                                | Hong Kong Monetary Authority (HKMA)                      |
|           |                                                |                                       |                                                | Financial Services Agency (FSA)                          |





Table 7.4 continued

| Country      | Resolution authority                                      |                                           |                                                        |                                                                |                                        |
|--------------|-----------------------------------------------------------|-------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------|----------------------------------------|
|              | Central bank and banking supervisor                       | Banking supervisor                        | Deposit insurance and bank resolution agency           | Integrated financial regulator                                 | Public agency/ government authority    |
| Korea        |                                                           |                                           | Korea Deposit Insurance Corporation (KDIC)             | Financial Services Commission<br>Financial Supervisory Service |                                        |
| Mexico       |                                                           |                                           | Instituto para la Protección al Ahorro Bancario (IPAB) |                                                                |                                        |
| Netherlands  | De Nederlandsche Bank (DNB)                               |                                           |                                                        |                                                                | Dutch Ministry of Finance <sup>1</sup> |
| Russia       | Central Bank of The Russian Federation (CBR)              |                                           | Deposit Insurance Agency                               |                                                                |                                        |
| Saudi Arabia | Saudi Arabian Monetary Agency (SAMA)                      |                                           |                                                        |                                                                |                                        |
| Singapore    |                                                           |                                           |                                                        | Monetary Authority of Singapore (MAS)                          |                                        |
| South Africa | Registrar of Banks, within South Africa Reserve Bank      |                                           |                                                        |                                                                |                                        |
| Spain        | Bank of Spain                                             |                                           | Fondo de reestructuración ordenada bancaria (FROB)     |                                                                |                                        |
| Switzerland  |                                                           |                                           |                                                        | Swiss Financial Market Supervisory Authority (FINMA)           |                                        |
| Turkey       |                                                           | Banking Regulation and Supervision Agency | Savings Deposit Insurance Fund                         |                                                                |                                        |
| UK           | Bank of England (BoE)                                     |                                           |                                                        |                                                                | HM Treasury <sup>2</sup>               |
| US           | Federal Deposit Insurance Corporation (FDIC) <sup>3</sup> |                                           |                                                        |                                                                |                                        |

Notes: <sup>1</sup> The Dutch Ministry of Finance is given resolution powers if the failure of a financial institution (or its parent firm) presents an immediate and serious threat to financial stability in the Netherlands.

<sup>2</sup> The UK HM Treasury is the relevant authority only in the case of temporary public ownership.

<sup>3</sup> The OCC (Office of the Comptroller of the Currency) may appoint a receiver for a federally licensed branch of a foreign bank.

Source: Collated from 'Thematic review on resolution regimes', Financial Stability Board (2013a).

In 2014, the FSB adopted additional guidance that elaborates on specific key attributes (KAs) relating to information sharing for resolution purposes and sector-specific guidance that sets out how the KAs should be applied for insurers, financial market infrastructures (FMIs) and the protection of client assets in resolution (FSB, 2014). The KAs set out the core elements of effective resolution regimes that apply to any financial institution that could be systemically significant or critical if it fails. Their implementation should allow authorities to resolve financial institutions in an orderly manner, without taxpayer exposure to loss from solvency support, while maintaining continuity of their vital economic functions. They set out essential features in 12 areas that should be part of the resolution regimes of all jurisdictions, which relate to:

- 1 scope of the resolution regime;
- 2 resolution authority (existence, mandate and governance);
- 3 resolution powers;
- 4 legal framework governing set-off rights, contractual netting, collateralisation arrangements and segregation of client assets;
- 5 the existence of safeguards;
- 6 funding arrangements to support the resolution of firms;
- 7 legal framework conditions for cross-border co-operation;
- 8 crisis management groups (CMGs);
- 9 institution-specific cross-border co-operation agreements (COAGs) – these mainly apply to global systemically important financial institutions (G-SIFIs);
- 10 resolvability assessments;
- 11 recovery and resolution planning;
- 12 access to information and information sharing.

#### 7.4.4 Co-operation and resolution processes

Finally, the viability of a safety net in case of crisis also depends on efficient communication and co-operation between all network elements (see Figure 7.2). This requires appropriately standardised and clearly regulated processes that at best are also internationally harmonised. The 2007–2009 global financial crisis highlighted the need for further work in this area, as information exchange and a co-ordinated decision-making process have proven difficult at the national level, let alone internationally.

## 7.5 Limitations of regulation

So far, we have highlighted the case for financial regulation, which depends mainly on various market imperfections and failures (information asymmetries, agency problems, etc.), which, in the absence of regulation, would produce sub-optimal results and reduce consumer welfare. As a consequence, the purpose of regulation should be limited to correcting for identified market imperfections and failures. There are, however, a number of arguments against regulation.

Regulatory arrangements, in particular the ‘safety net’ arrangements create moral hazard. The concept of moral hazard was introduced in Chapter 1. Deposit insurance and the LOLR can cause people to be less careful than they would be otherwise. For example, with 100 per cent deposit insurance, depositors will not be concerned about the behaviour of their bank. Similarly, the belief that the LOLR will eventually bail out troubled banks may encourage institutions to take greater risks in lending. Box 7.2 illustrates these concepts. Other examples of the moral hazard caused by the **government safety net** are known as the **too big to fail** (TBTF) and the **too important to fail** (TITF) cases. Because the failure of a large (or strategically important) bank poses significant risks to other financial institutions and to the financial system as a whole, policy makers may respond by protecting bank creditors from all or some of the losses they otherwise would face. If managers of large (or important) banks believe that they will be bailed out by the authorities (with taxpayers’ money) if they get into financial difficulty then this increases the moral hazard incentives for big banks, resulting in banks taking on even greater risks to increase profits and executive remuneration. Executives may be less concerned about taking big risks if they know depositors and the bank will ultimately be protected by government bailouts if things go wrong.

### BOX 7.2 MORAL HAZARD AND GOVERNMENT SAFETY NET ARRANGEMENTS

Financial regulation and supervision are needed because moral hazard can be associated with government safety net arrangements that are designed to protect the banking and financial system. For example, banks that face liquidity problems and cannot borrow from other banks in the market may approach the regulators to act as a lender of last resort in order to provide emergency liquidity assistance. This, in principle, seems a good thing as the authorities have a mechanism for providing liquidity to the banking system at times of crises. However, moral hazard arises in that if banks all believe they have access to the LOLR, they may be inclined to take on excessive risks knowing that in the event of trouble they will be bailed out by the authorities (in other words, the taxpayer, as these are public funds being used). To mitigate this moral hazard problem the authorities need to establish a regulatory framework that assures that access to the LOLR facility is by no means guaranteed for banks.

Linked to this is the too big to fail argument, whereby the largest banks are viewed as being too big to be allowed to fail and therefore they must have guaranteed access to the LOLR – which could cause moral hazard problems. No financial regulatory authority will ever provide guaranteed access to the LOLR financing, although history does tell us that (for systemic and other reasons) large banks are likely to be bailed out more than small banks. In addition, the too important to fail view argues that size by itself is not the relevant criteria for bailing out troubled banks – rather the significance or importance of banks in specific markets and the expected scale/impact of potential failure should be the main criteria in providing support.

Similar moral hazard issues relate to the design of appropriate deposit insurance and other investment (and insurance) compensation schemes. As noted earlier, if deposit insurance is too generous it creates incentives for banks to take on more risk as they know their customers’ deposits will be protected in the event of a bank failure. Similar arguments can be put forward for other compensation schemes. Financial regulations need to be designed to reduce such possible moral hazards occurring; it is important to note that regulation can never eliminate all information asymmetries, but it can (and should) be formulated in order to minimise the potential adverse effects of such market failures.

Banks may also benefit from **regulatory forbearance**. Regulatory forbearance (or renegotiation) is an example of time inconsistency. Time inconsistency refers to the problem that it may not be optimal *ex post* (after an event occurs) to implement regulations that were optimal *ex ante* (before the event occurred). When financial intermediaries are in trouble, there may be pressures not to apply existing regulations, for example, to impose higher capital or liquidity requirements. This is because it could worsen the institution's problems. If the bank was allowed to fail, this could drain the deposit insurance fund. Furthermore, publicity surrounding a bank facing difficulties may worry the public, who may be induced to withdraw their savings, thereby aggravating the bank's problems (with a possible domino effect on other institutions leading to further bank failures). Also, there might be political costs associated with enforcing regulations and therefore an incentive to delay action. There are some benefits of forbearance. First, not publicising the bank's problems may help avoiding systemic risk caused by bank runs. In addition, the bank may be worth more as a 'going concern', that is, remaining in operation rather than going out of business and liquidating its assets. To stay operational, a bank must be able to generate enough resources. As we have seen, banks assets are highly illiquid and therefore their sale might not generate enough cash to satisfy creditors.

There are, however, costs associated with forbearance. First, it may cause moral hazard: forbearance in one case may lead to expectations of similar behaviour in future cases, causing other financial institutions to observe regulations less carefully. Furthermore, regulators and regulated firms may become locked into an ever-worsening spiral, resulting in a loss of public confidence in how banks and the financial system in general are being regulated.

Regulation can create problems of **agency capture** – that is, the regulatory process can be 'captured' by producers (in this case by banks and other financial institutions) and used in their own interest rather than in the interests of consumers. For example, some have argued that the Basel II Capital Accord had too much input from banking sector participants and large banks in particular. The Basel II capital rules allowed the largest banks to use their own internal models for assessing risk and capital adequacy positions – which are likely to have led the biggest banks to hold less capital for regulatory purposes (see also Section 7.10). The fact that major banks have had a strong say in devising regulations that govern their own operations is a possible indicator of agency capture.

Regulation is a costly business and the costs of compliance with the regulatory process will be passed on to consumers, resulting in higher costs of financial services and possibly less intermediation business. In addition, regulatory costs may act as a barrier to entry in the market and this may consolidate monopoly positions.

The notion of incremental **compliance costs** is set out in Alfons and Andrews (1999), p. 16 as follows:

Compliance costs are the costs to firms and individuals of those activities required by regulators that would not have been undertaken in the absence of regulation. Thus the term 'compliance costs' as used here refers to the incremental costs of compliance caused by regulation, not to the total cost of activities that happen to contribute to regulatory compliance. Examples of compliance costs include the costs of any additional systems, training, management time and capital required by the regulator.

However, none of these criticisms are enough to reject financial regulation. Regulation is always about making judgements and considering trade-offs between costs and benefits. While it is important to recognise the limitations of regulation, a well-designed regulatory framework is necessary to ensure consumers' confidence in the financial sector. Although there are costs involved, there is also evidence that consumers and other users demand appropriate regulation: public pressure to introduce regulations may derive from the view that market solutions to regulations do not provide users with the appropriate reassurance that they are being protected appropriately.

### 7.6 Causes of regulatory reform

The scope and complexity of financial regulation has tended to grow almost continually. This has been partially in response to public reaction to financial scandals and the consequent political pressures generated (i.e. increased consumers demand for regulation). Another factor that can change regulations is financial innovation. As new financial products and services emerge and gain in market significance, there are often calls for new regulation – for instance, the US Federal Reserve in early 2005 called for greater regulation of hedge funds (which are private investment funds that trade and invest in various assets such as securities, commodities, currency and derivatives on behalf of their clients) due to their rapid growth and potentially destabilising activity. Similar examples can be given regarding the regulation of derivatives activity and other financial instruments. The global financial crisis of 2007–2009 attracted regulatory attention to innovations such as derivatives, off-balance-sheet vehicles, hedge funds, private equity firms, etc.

One of the reasons financial innovations tend to attract regulatory attention is that often the innovations are due to regulatory avoidance. In other words, financial firms and markets create new products not only to meet new demands but also to circumvent regulations. For example, the growth of off-balance-sheet activities (derivatives trading, securities underwriting, foreign exchange trading, and so on) during the first half of the 1980s can be explained by the fact that this business was not subject to capital regulations, in contrast to on-balance-sheet business. Similarly, the flow of US dollars to the United Kingdom during the 1960s and the start of the dollar Eurobond market has been explained mainly because restrictive regulations in the United States (reserve requirements, limits on deposit rates – Regulation Q, and other limits on United States domestic bond issues) encouraged dollars to flow to London and borrowers to raise dollar debt finance by issuing Eurobonds. Firms innovate to get around regulations and the regulators are always one step behind the market – this is known as the regulatory dialectic.

As mentioned earlier, other factors impacting on regulatory reform are internationalisation and globalisation trends. The increased international activity of financial firms means that foreign institutions play an increasing role in many domestic financial sectors. Throughout the world, financial liberalisation has provided a passport for banks to offer services cross-border.

The increased presence of foreign financial firms raises issues relating to how they should be regulated. The main issue relates to who is ultimately responsible if a foreign bank faces difficulties in an overseas market – should it be the host or home country regulator? Generally, for large complex banks, the host regulator will supervise foreign subsidiary activity

but it is the home country that is ultimately responsible if the bank faces difficulties (see also Chapter 4, where these issues are discussed in greater detail).

In addition to the issue of regulatory responsibility, the internationalisation trend has also encouraged much greater debate about convergence of rules – so as to ensure that banks operate under similar regulations in different jurisdictions. It has been argued that minimal harmonisation (regulation based on minimum standards) allows for greater flexibility in implementing legislation and is likely to result in (or sustain) more competitive playing fields than if one chooses maximum harmonisation. Harmonisation should always result in some form of convergence in national rules and at the same time should increase actual and potential competition (as well as the safety of the system) if it is to be effective. Many commentators recognise the need for greater co-operation between supervisory authorities and improved relations between supervisory authorities, market participants and consumers.

Another factor impacting on regulatory reform – and closely linked to the internationalisation trend – is the globalisation phenomena. The growth in international activities and trade of multinational corporations has increased the demand for services from financial institutions that operate cross-border and therefore financial firms continue to expand their international presence. This means that various financial firms operate globally (e.g. HSBC, Santander, Citibank, Goldman Sachs, etc.). As a result, banks are increasingly exposed to risks originating from abroad, and risks to financial stability are less and less confined to national borders. This calls, at the minimum, for greater regulatory oversight and co-ordination between national regulators. Further, consolidation in the global banking industry has resulted in the emergence of financial conglomerates that conduct an extensive range of businesses with a group structure. The formation of financial conglomerates is forcing regulators to re-think the way in which the financial sector should be supervised.

In addition to the aforementioned factors, various other forces can have a marked impact on the regulatory environment. Major financial crises can have a big impact on regulatory changes, mainly because the occurrence of a crisis is an indication that regulation in place prior to the difficulties was not sufficient. However, not all countries tightened regulatory restrictions following the global financial crisis. A survey carried out by Barth *et al.* (2013) showed that 80 per cent of countries tightened such restrictions following the crisis, while others actually eased overall restrictions, including Brazil, Portugal and Switzerland.

One of the mainstays of banking sector policy around the world is capital regulation. Many rules and policies determine the precise amount and nature of capital that banks must hold. These are analysed in detail in Section 7.7.

## 7.7 Bank capital regulation

The role of capital in the financial sector, and for banks in particular, is a central element of regulation. A bank's capital may be defined as the value of its net assets (i.e. total assets minus total liabilities). In practice, this capital is the sum of the bank's paid-up share capital and its accumulated capital reserves. A bank's capital is vital for the protection of its depositors, and hence for the maintenance of general confidence in its operations, and the underpinning of its longer-term stability and growth.

The role of capital in banking can be illustrated by a simple balance sheet diagram as shown in Table 7.5, where Bank Greedy has assets of £55 billion and £5 billion of capital. The bank has £54 billion in loans and £50 billion held in deposits.

**Table 7.5** Bank Greedy balance sheet

| Liabilities (£) |            | Assets (£)             |            |
|-----------------|------------|------------------------|------------|
| Capital         | 5 billion  | Cash and liquid assets | 1 billion  |
| Deposits        | 50 billion | Loans                  | 54 billion |
| Total           | 55 billion | Total                  | 55 billion |

Now let us assume that the bank has made some risky loans and £4 billion-worth of loans go bad. The bank believes they will never be repaid (the bank cannot recover these loans). Bank Greedy has to take a 'hit' and the losses can be borne by the capital cushion. As shown in Table 7.6, assets shrink by £4 billion and capital falls to £1 billion.

**Table 7.6** Bank Greedy balance sheet after £4 billion loans go bad

| Liabilities (£) |            | Assets (£)             |            |
|-----------------|------------|------------------------|------------|
| Capital         | 1 billion  | Cash and liquid assets | 1 billion  |
| Deposits        | 50 billion | Loans                  | 50 billion |
| Total           | 51 billion | Total                  | 51 billion |

In this case Bank Greedy can bear the loss of £4 billion as it has sufficient capital to cover these losses. Note that we assume that cash and liquid assets remain at £1 billion. If, however, the losses exceed £5 billion then Bank Greedy does not have enough capital to cover these losses and it cannot meet depositor obligations. See what happens if instead of a £4 billion loss Bank Greedy has £7 billion loans go bad. This is shown in Table 7.7.

**Table 7.7** Bank Greedy balance sheet after £7 billion loans go bad

| Liabilities (£) |            | Assets (£)             |            |
|-----------------|------------|------------------------|------------|
| Capital         | 0 billion  | Cash and liquid assets | 1 billion  |
| Deposits        | 48 billion | Loans                  | 47 billion |
| Total           | 48 billion | Total                  | 48 billion |

It can be seen that Bank Greedy has used all its £5 billion capital to cover these losses and deposits of £2 billion have also had to be used to make up the shortfall – this means that the bank cannot repay all its depositors as the value of deposits has fallen from the original £50 billion to £48 billion – in theory the bank would have to tell its depositors that it had some bad news and unfortunately some will not be able to withdraw their deposits (or that all depositors will bear a loss). Of course, in reality this does not happen as the bank is insolvent.

The main point to stress is that any losses incurred by a bank – whether these are caused by bad loans, securities trading, the failure of a subsidiary, fraudulent activity or whatever – have to be met out of its capital as deposits have to be protected at all costs in order to maintain confidence in the bank, as well as the banking system overall. This is why bank regulators spend so much



time and energy focusing on the capital adequacy of banks. The adequacy of any given amount of capital not only depends upon the absolute volume of assets to be covered, but is also affected by the quality of those assets. The more risky the assets, the greater must be the cushion of capital funds, all other things being equal, in order to maintain a given level of capital adequacy.

For a number of years, the Bank of England specified for each UK bank individually a minimum required ratio of capital to risk-weighted assets. If the actual ratio were to fall below this 'trigger' ratio, the Bank of England would be likely to intervene in the bank's activities. This regulatory capital ratio was set to take account of the Bank of England's assessment of the bank's managerial capacity with regard to its risk position, its profitability and its overall prospects. In addition, it was expected that, in normal circumstances, each bank would maintain a 'target' capital ratio that included a margin over the value of its trigger ratio. The UK banking sector regulator has modified this approach in the light of evolving international standards. Moves at an international level to harmonise the capital adequacy ratios of banks in different countries have led to a more rigorously defined framework for capital adequacy. The Committee on Banking Regulations and Supervisory Practices of the Bank for International Settlements (BIS) put forward a framework in July 1988 for the harmonisation of standards of capital adequacy. This framework has become known as the Basel I Accord. The objective of this framework was to strengthen the world's banking system and place it in a better position to withstand any future problems in world financial markets. In addition, the requirements were intended to provide a more equal basis for competition between banks in different countries and to remove the incentive for banks to relocate activities to other countries in order to take advantage of relatively lax regulatory requirements.

Capital adequacy ratios are a measure of the amount of a bank's capital expressed as a percentage of its risk-weighted credit exposures. The Basel I Accord recommends minimum capital adequacy ratios to ensure banks can absorb a reasonable level of losses before becoming insolvent. Applying minimum capital adequacy ratios serves the purpose of protecting depositors and promoting the stability and efficiency of the financial system. As such, minimum capital standards can be seen as a vital tool to reducing systemic risk. Although the Basel Committee does not have any powers to impose the Accord, more than 100 countries have implemented its guidelines in one form or another.

## 7.8 The 1988 Basel Capital Accord (Basel I)

The Basel Committee on Banking Supervision was created at the end of 1974 and was charged by the Group of Ten (or G-10, the largest ten industrialised countries in the world: Belgium, Canada, France, Germany, Italy, Japan, Netherlands, Sweden, the United Kingdom and the United States), plus Luxembourg and Switzerland, central bank governors to seek a common approach among its members towards measuring capital adequacy and the prescription of minimum capital standards.<sup>6</sup> In July 1988, the Basel Committee on Banking Supervision introduced its 1988 Capital Accord (the Basel I Accord). The majority of the world's leading central banks undertook to implement the Basel Accord by the end of 1992. (The EU implemented almost all of the features of the Basel Accord into EU and therefore member states law by the end of 1992.)

<sup>6</sup>The Basel Committee is the Committee of Central Banks and Bank Supervisors from the major industrialised countries that meets every three months at the BIS in Basel, Switzerland. Note the French spelling of the name of the Swiss city is Basle, whereas the German spelling is Basel. Nowadays the latter is more commonly used.

The Accord reflected the culmination of the committee work on international convergence of capital adequacy and is based on a risk–asset ratio (RAR) approach (see Box 7.3). The committee stated that ‘a weighted risk ratio in which capital is related to different categories of asset and off-balance-sheet exposure, weighted according to broad categories of relative riskiness, is the preferred method for assessing the capital adequacy of banks’ (BCBS, 1988). The 1988 Capital Accord established an international standard around a capital ratio of 8 per cent and focused on risks associated with lending (credit risks) thereby ignoring other types of risk. The Basel I definition of capital is made up of two elements: Tier 1 (‘core capital’) and Tier 2 (‘supplemental capital’). Bank total capital is the sum between Tiers 1 and 2 (‘capital base’).

Specifically, the elements of capital are as follows:

- *Tier 1*
  - (a) Ordinary paid-up share capital/common stock
  - (b) Disclosed reserves
- *Tier 2*
  - (a) Undisclosed reserves
  - (b) Asset revaluation reserves
  - (c) General provisions/general loan loss reserves
  - (d) Hybrid (debt/equity) capital instruments
  - (e) Subordinated term debt

More details on the definitions of capital elements can be found in Box 7.3. The sum of Tier 1 and Tier 2 elements is eligible for inclusion in the capital base, subject to various

### BOX 7.3 DETAILS OF CAPITAL ELEMENTS (ESTABLISHED 1988 AND APPLIED IN 1992)

**Tier 1 (core capital)** = common stockholders’ equity + non-cumulative perpetual preferred stock + any surplus + minority interest in the equity accounts of consolidated subsidiaries – goodwill and other intangibles (deduction is carried out only if some conditions are met).

Required TIER 1 core capital is equal to risk weight  $\times$  4% of risk-weighted assets

**Tier 2 (supplementary capital)** = allowance for losses on loans and leases (reserves) + cumulative perpetual, long-term and convertible preferred stock + perpetual debt and other hybrid debt/equity instruments + intermediate-term preferred stock and term subordinated debt. The total of Tier 2 is limited to 100 per cent of Tier 1. Other limitations are specified in the 1992 revised guidelines.

Deductions from total capital (Tier 1 + Tier 2) consist of investments in unconsolidated banking and financial subsidiaries, reciprocal holdings of capital securities and other deductions (such as other subsidiaries or joint ventures) as determined by supervisory authorities with handling on a case-by-case basis or as a matter of policy after formal rule making.

Required total capital (TIER 1 + TIER 2 – deductions) is equal to risk weight  $\times$  8 per cent of risk-weighted assets



**Figure 7.4** Measures of bank capital

limits described in the 1988 Basel report. Figure 7.4 illustrates the different measures of bank capital.

The general framework for capital adequacy risk-weighted assets can be summarised as follows. There are four risk classes in the risk-weighted system that reflect credit risk exposure:

- 1 No risk: 0 per cent (e.g. cash or equivalents, bonds issued by OECD governments).
- 2 Low risk: 20 per cent (e.g. short-term claims maturing in a year or less; bonds issued by agencies of OECD governments).
- 3 Moderate risk: 50 per cent (e.g. mortgages).
- 4 Standard risk: 100 per cent (e.g. commercial loans; claims by non-OECD banks and government debts).

#### BOX 7.4 RAR (RISK-ASSET RATIO) APPROACH

The RAR is a relative simple approach that sets out to appraise capital adequacy on the basis of banks' relative riskiness. Banks' assets are divided by the supervisory authorities into a number of equivalent risk classes. Different 'risk-weights' are assigned to each of the equivalent risk classes of assets. Total risk-weighted assets are calculated as follows:

$$RWA = \sum a_i r_i \quad \text{where} \quad A = \sum a_i \quad \text{and}$$

$$RAR = \frac{C}{RWA} \quad \text{where} \quad RWA = \text{total risk-weighted assets}$$

RAR = risk-asset ratio  
 A = bank total assets  
 $a_i$  = risk classes of assets  
 $r_i$  = risk weights  
 C = capital as defined by the supervisory authorities

Moreover, conversion factors were set for calculating credit-equivalent amounts for off-balance-sheet (OBS) items:

- (a) 0 per cent (e.g. unused portion of loan commitments);
- (b) 20 per cent (e.g. commercial letters of credit);



Box 7.4 RAR (risk asset ratio) approach (*continued*)

- (c) 50 per cent (e.g. revolving underwriting facilities);
- (d) 100 per cent (e.g. standby letters of credit).

Essentially, the capital adequacy scheme is based on a four-step approach:

- 1 *Classify* assets into one of four risk categories described above.
- 2 *Convert* OBS commitments and guarantees on their on-balance-sheet 'credit equivalent' values and classify them in the appropriate risk category.
- 3 *Multiply* the £ amount of assets in each risk category by the appropriate risk weight; this equals 'risk-weighted assets'.
- 4 *Multiply* 'risk-weighted assets' by the minimum capital percentages, either 4 per cent for tier 1 capital or 8% for total capital for a bank to be adequately capitalised.

For example, take a bank with the following assets:

- cash: \$100m (0 per cent risk weighting);
- loans to other banks: \$500m (20 per cent risk weighting);
- mortgage loans to owner-occupiers: \$800m (50 per cent risk weighting);
- commercial loans: \$1,500m (100 per cent risk weighting).

The regulatory minimum capital ratio can be calculated as follows:

Total risk-weighted assets (RWA) =  $(\$100m \times 0) + (\$500m \times 0.2) + (\$800m \times 0.5) + (\$1,500m \times 1) = \$0 + \$100m + \$400m + \$1,500m = \$2,000m$

The minimum capital requirement is 8 per cent of \$2000m = \$160 million, of which at least 50% needs to be held in the form of Equity (Tier 1) capital.

If the RAR calculated by the bank falls below the minimum ratio stipulated by the regulatory authorities then this obviously indicates the institution has inadequate capital. A summary of the minimum capital adequacy requirements according to the Basel I accord, and a guide as to how to distinguish well-capitalised banks from those that are undercapitalised, is given in Box 7.5.

The Basel I Accord was generally regarded as a step forward in the regulation of bank capital adequacy. It involved international agreement and it became the basis for most nations' capital regulations for all banks. Nevertheless, almost immediately debate began as to its efficiency and effectiveness. Questions were raised on capital ratios appearing to lack economic foundation, risk weights not accurately reflecting the risk associated with assets (e.g. the riskiness of loans) and the lack of recognition of asset portfolio diversification. It should also be noted that many nations chose to set capital adequacy ratios somewhat higher than the Accord's minimum, reflecting their own assessment of the risk associated with individual banks' activities.

## BOX 7.5 FIVE CAPITAL-ADEQUACY CATEGORIES OF BANKS

1 *Well-capitalised:*

- Total capital to risk-weighted assets 10 per cent
- Tier 1 capital to risk-weighted assets 6 per cent
- Tier 1 capital to total assets 5 per cent

2 *Adequately capitalised (fulfilling minimum requirements):*

- Total capital to risk-weighted assets 8 per cent
- Tier 1 capital to risk-weighted assets 4 per cent
- Tier 1 capital to total assets 4 per cent

3 *Undercapitalised:*

- Fails to meet one or more of the capital minimums for an adequately capitalised bank

4 *Significantly undercapitalised:*

- Total capital to risk-weighted assets <6 per cent
- Tier 1 capital to risk-weighted assets <3 per cent
- Tier 1 capital to total assets <3 per cent

5 *Critically undercapitalised:*

- $$\frac{[(\text{Common equity capital} + \text{Perpetual preferred stock} - \text{Intangible assets}) / \text{Total assets}]}{\leq 2 \text{ per cent}}$$

## 7.9 The 1996 amendments to the 1988 Accord

The original capital Accord focused on risks arising from the lending activity of banks thereby ignoring other types of risks (for an overview of the main bank risks see Chapter 11). A proposal for changing the original Accord to include capital charges for market risk incurred by banks was issued by the Basel Committee in April 1995. The objective was to provide ‘an explicit capital cushion for the price risks to which banks are exposed, particularly those arising from trading activities’. The Basel Committee defined market risk as the risk of losses in on- and off-balance-sheet positions arising from movements in market prices.

In particular, the risks covered by the proposed framework were: (a) the bank’s trading book position in debt, equity instruments and related off-balance-sheet contracts and (b) commodity and foreign exchange positions held by the bank. The amendments to the Accord to incorporate market (trading) risk resulted in the inclusion of an ‘ancillary’ or Tier 3 capital to support trading book activities. Moreover, a significant innovation in the market risk requirements consisted in the opportunity given to banks to use their internal risk-assessment models (see Chapter 12 for more details) for measuring the riskiness of their trading portfolios.

Overall, these regulatory changes in the minimum capital required for market risk came into force in 1996, and represented a significant step forward for the Committee in strengthening the soundness and stability of the international banking and financial systems.

## 7.10 The second Capital Accord (Basel II)

In response to the criticisms of the original Accord, a number of changes were made. On 3 June 1999, the Basel Committee on Banking Supervision (BCBS) formally launched proposals for a new capital adequacy framework (which was to become known as Basel II). The first version of the proposed Accord received strong criticism from bankers and academics, which prompted the committee to make substantial changes to the new framework. In May 2004, central bank governors and the heads of bank supervisory authorities in the Group of Ten (G10) endorsed the publication of the 'International convergence of capital measurement and capital standards: A revised framework' (BCBS, 2004).

In June 2006, the Committee released a comprehensive version of the Accord, incorporating the June 2004 Basel II Framework, the elements of the 1988 Accord that were not revised during the Basel II process, the 1996 amendment to the Capital Accord to incorporate Market Risks, and the 2005 amendments (BCBS, 2006). This was the outcome of the Basel Committee's work over the years to secure international convergence on revisions to supervisory regulations governing the capital adequacy of internationally active banks. The Accord's main aim was to introduce a more comprehensive and risk-sensitive treatment of banking risks. In particular, the setting of minimum capital requirements was based on an update of the risk-weighting approach including the use of banks' internal risk ratings and external credit risk assessments. Table 7.8 summarises the key differences between Basel I and Basel II, highlighting the main reasons put forward to revise the first capital accord.

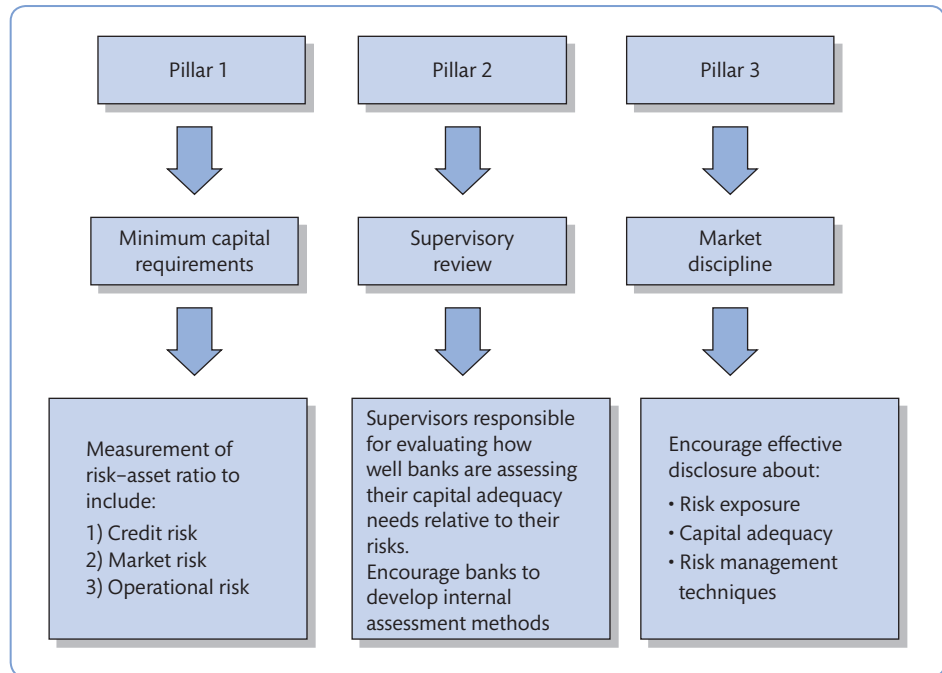
Basel II was built on three main pillars. Pillar 1 dealt with the quantification of new capital charges and relied heavily on banks' internal risk-weighting models and on external rating agencies. Pillar 2 defined the supervisory review process and Pillar 3 focused on market discipline, imposing greater disclosure standards on banks in order to increase transparency. Figure 7.5 summarises the three-pillars approach.

### Pillar 1

The first pillar sought to amend the old rules by introducing risk weightings that were more closely linked to borrower's credit standing. Basel II refined the methodology to reflect with greater precision the varying underlying risks against which banks were required to hold

**Table 7.8** Rationale for Basel II: more flexibility and risk sensitivity

| Basel I                        | Basel II                                                                                    |
|--------------------------------|---------------------------------------------------------------------------------------------|
| Focus on a single risk measure | More emphasis on banks' on internal methodologies, supervisory review and market discipline |
| One size fits all              | Flexibility, menu of approaches, incentives for better risk management                      |
| Broad brush structure          | More risk sensitivity                                                                       |



**Figure 7.5** Basel II: the three pillars

capital. The second Accord did not change the definition of capital or the minimum requirement of 8 per cent capital to risk-weighted assets. It mainly affected how banking risks (credit risk, which is risk of a borrower's default, operational risk, which is the risk associated with the potential for systems failure, and market risk, which is the risk that the value of investments will decrease due to movements in market factors) were measured. The biggest changes relate to the calculation of capital backing for credit risk. Under Basel II, banks would be able to choose from what is known as the 'menu of approaches', including the 'standardised approach' and the internal ratings based (IRB) approach. In the standardised approach, Basel II defined risk weights within broad categories of sovereigns, banks and companies, by reference to an external credit assessment firm (credit rating agency) subject to strict standards. Under IRB, banks were allowed to use their internal credit risk assessments subject to strict methodological and disclosure requirements.

### **Pillar 2**

Pillar 2 identified the roles of the national supervisors to ensure banks use appropriate methodology to determine capital adequacy ratios and have a strategy to maintain such ratios. These were defined as follows:

- to review banks' internal assessment procedures and strategies, taking appropriate action if these fall below standard;
- to encourage banks to hold capital above the minimum requirements;
- to intervene as early as possible to ask banks to restore their capital levels if they fall below the minimum.



The supervisory review process aimed to ensure that a bank's capital adequacy position is consistent with its overall risk profile. To this end, bank regulators must be able to make qualitative judgements on the ability of each bank to measure and manage its own risks. Supervisors should also have the ability to require banks to hold capital in excess of minimum regulatory requirements.

### Pillar 3

The third pillar sought to enhance effective market discipline by introducing high disclosure standards with regard to bank capital. This required banks to provide more reliable and timely information, enabling market participants to make better risk assessments. Banks were expected to disclose:

- risk exposure;
- capital adequacy;
- methods for computing capital requirements;
- all material information.

Taken together, Pillar 1 provided the rules for quantifying risk sensitivity and the minimum capital charges associated with these risks. This was balanced by the supervisory judgments available under Pillar 2 and market disclosure rules of Pillar 3. Ultimately, the Basel II Accord aimed to create a more comprehensive and flexible regulatory framework, without sacrificing the safety and soundness achieved by the Basel I Accord.

### 7.10.1 Criticisms of Basel II

A number of early criticisms were made about Basel II. They can be summarised as follows:

- 1 It is pro-cyclical, i.e. it moves with the economic cycle. The literature on the pro-cyclical effects of Basel II suggested that the rules may exacerbate downturns in the business cycle. For instance, as credit risk increases in a recession, capital requirements are likely to rise, inducing credit rationing.
- 2 Basel II could increase the amount of systemic risk for banks using the standardised approach. These banks have little incentive to diversify, as they are not rewarded for it.
- 3 Some suggested that there was a danger of banks that are part of a financial conglomerate moving their credit risk to another non-bank financial subsidiary to reduce the amount of capital they have to set aside.
- 4 Another concern is that Basel II requirements could encourage banks to transfer credit risk off their balance sheets (see Chapter 18 on the use of asset-backed securitisation and the use of financial derivatives).
- 5 Small and medium-size enterprises and firms located in developing countries may find it difficult to raise finance, as they are not listed and therefore do not have official credit ratings.
- 6 There is a potential conflict of interest as rating agencies also act as advisors to big banks on their risk management systems.

Some of these early criticisms were brought to the fore by the 2007–2009 global financial crisis. Since then, a number of amendments were proposed. Responses to the global financial crises called for tighter and 'more global' regulation of financial institutions. Some doubts, however, remain as to whether the attempts to regulate globally (i.e. Basel II) have if not caused then worsened the aforementioned 2007–2009 crisis.

## 7.11 The third Capital Accord (Basel III)

Basel II, to all intents and purposes, never came fully into effect. The global financial crisis laid bare the shortcomings of the existing prudential framework and made a thorough overhaul an overriding necessity. In July 2009, less than two years after Basel II came into force in the EU, the Basel Committee complemented its Basel II rules on trading books (also known as 'Basel 2.5').<sup>7</sup> The G20 approved the new Basel III solvency and liquidity rules at its Seoul summit in November 2010.

In December 2010 and June 2011, the BCBS published its latest recommendations on bank solvency and liquidity (BCBS, 2011a).

According to the BCBS, 'Basel III' is a comprehensive set of reform measures, developed by the Basel Committee on Banking Supervision, to strengthen the regulation, supervision and risk management of the banking sector. These measures aim to:

- improve the banking sector's ability to absorb shocks arising from financial and economic stress, whatever the source;
- improve risk management and governance;
- strengthen banks' transparency and disclosures.

The reforms target:

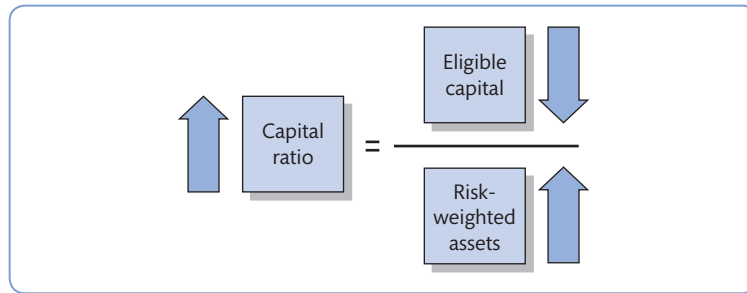
- bank-level, or micro-prudential, regulation, which will help raise the resilience of individual banking institutions to periods of stress;
- macro-prudential, system-wide risks that can build up across the banking sector as well as the pro-cyclical amplification of these risks over time.

The proposal was first published in December 2010, with a revised version issued in June 2011. In addition, the Basel Committee has issued the full text of the revised liquidity coverage ratio (LCR) in January 2013 (BCBS, 2013a). Being a new and complex set of rules, in the remainder of this section we will introduce the main changes under the new regulatory standards, as well as the timetable that has been set forth for their implementation.

The Basel III Accord addresses several issues. It proposes many new capital, leverage and liquidity standards to strengthen the regulation, supervision and risk management of the banking sector. The capital standards and new capital buffers will require banks to hold more capital and higher quality capital than under Basel II. The new leverage and liquidity ratios introduce a non-risk-based measure to supplement the risk-based minimum capital requirements and measures to ensure that adequate funding is maintained in case of crisis.

The enhanced capital ratios prescribed by the BCBS relate to the ratio of a firm's eligible regulatory capital divided by a regulatory prescribed calculation of risk-weighted assets. As illustrated in Figure 7.6, all three parts of this have changed, putting more pressure on a firm's compliance with the ratio. The capital ratio requirement has increased; the eligibility of

<sup>7</sup> The European Union implemented the Basel II Accord via the following EU Capital Requirements Directives: Directive 2006/48/EC of 14 June 2006 relating to the taking up and pursuit of the business of credit institutions and Directive 2006/49/EC of 14 June 2006 on the capital adequacy of investment firms and credit institutions.



**Figure 7.6** Elements of the capital ratio affected by Basel III

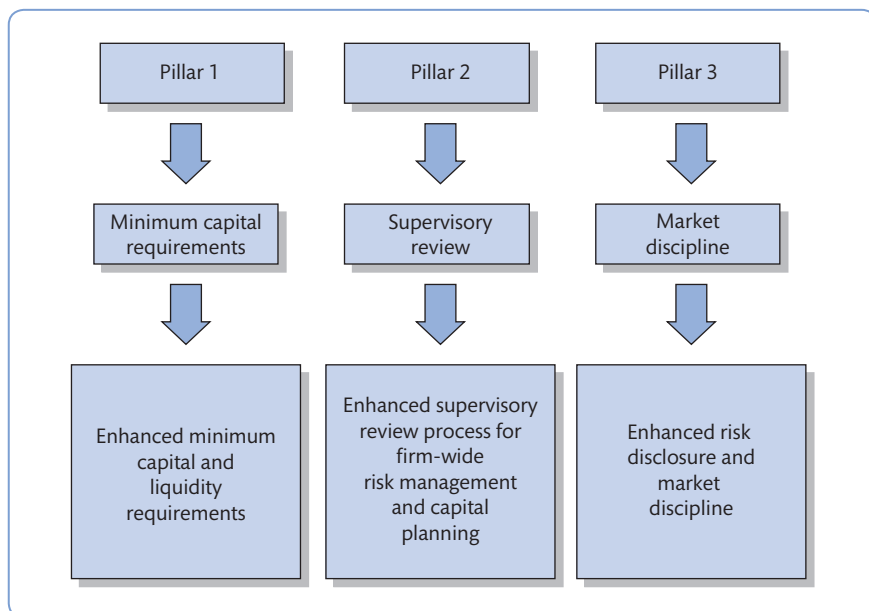
capital has been tightened so reducing the types of capital firms can use to meet the required ratio; and the calculation of risk-weighted assets has changed leading to an increase for many organisations.

Basel III builds upon the Basel II three pillars approach, and strengthens the three pillars, especially Pillar 1 with enhanced minimum capital and liquidity requirements. Figure 7.7 illustrates the key changes compared to Basel II.

### 7.11.1 The key elements of Basel III

The Basel III regulations aim to raise the quality, consistency and transparency of banks' capital base. Alongside the higher capital requirements and increased capital ratios, Basel III also introduces liquidity and leverage ratios. The main elements of the Basel III capital framework are:

- higher minimum Tier 1 capital requirement;
- a capital conservation buffer;



**Figure 7.7** Basel III: enhancing the three-pillar approach

- a countercyclical capital buffer;
- higher minimum Tier 1 common equity requirement;
- minimum total capital ratio.

### *Higher minimum Tier 1 capital requirement*

The crisis revealed that certain Tier 1 capital instruments – classed as core capital – were unable to absorb losses. To address this issue, Basel III has therefore tightened its definitions of regulatory capital. Tier 1 capital will now be comprised predominantly of common equity and retained earnings. Common equity will now reach 82.3 per cent of Tier 1 capital, inclusive of the capital conservation buffer. Basel III abandons the ‘core Tier 1’ concept in favour of the stricter ‘common equity Tier 1’. The proposed requirements include increases in Tier 1 capital ratio from 4 per cent to 6 per cent. The proposed timetable for the adoption of the higher minimum Tier 1 ratio was set at 4.5 per cent from 1 January 2013, 5.5 per cent from 1 January 2014 and 6 per cent from 1 January 2015. However, the implementation date has been postponed several times, and banks now have until 1 January 2022 to implement the changes. Nonetheless, the ‘Eighteenth progress report on adoption of the Basel regulatory framework’ (BIS, 2020b) shows that most jurisdictions have implemented the higher Tier 1 requirements.

### *A capital conservation buffer*

A capital conservation buffer of 2.5 per cent is to be used to absorb losses during periods of financial and economic stress. This requirement, which is designed to help banks withstand future periods of stress, will bring the total common equity requirement to 7 per cent. This is the sum of 4.5 per cent common equity requirement and the 2.5 per cent capital conservation buffer. It must be noted that the capital conservation buffer must be met exclusively with common equity and banks that do not maintain the capital conservation buffer will face restrictions on payouts of dividends, share buybacks and bonuses. Most large banks will likely maintain such a buffer, as falling below it will lead to additional regulatory scrutiny.

### *A countercyclical capital buffer*

Basel III also introduced a countercyclical buffer within a range of 0–2.5 per cent of common equity (or other fully loss-absorbing capital). This is in effect an extension of the conservation buffer and will be implemented according to national circumstances.

### *Higher minimum Tier 1 common equity requirement*

There is also an increase of Tier 1 common equity requirements from 2 per cent to 4.5 per cent. Common equity Tier 1 capital (CET1) is the highest quality of regulatory capital, as it absorbs losses immediately when they occur. Additional Tier 1 capital (AT1) also provides loss absorption on a going-concern basis, although AT1 instruments do not meet all the criteria for CET1. Table 7.9 summarises the components of regulatory capital.

### *Minimum total capital ratio*

The minimum total capital ratio will remain at 8 per cent. As illustrated in Table 7.9, the addition of the capital conservation buffer increases the total amount of capital a bank must hold to 10.5 per cent of risk-weighted assets. In the EU, the Capital Requirements Directive (2013/36/EU) (CRD IV) implemented the Basel III agreement.<sup>8</sup> At the time of writing (May 2021), the

**Table 7.9** Components of regulatory capital

|        |                             |                                                                                                                                                                                           |                          |
|--------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Tier 1 | Common equity Tier 1 (CET1) | Sum of common shares (equivalent for non-joint stock companies) and stock surplus, retained earnings, other comprehensive income, qualifying minority interest and regulatory adjustments | CET1 > 4.5%              |
|        | Additional Tier 1 (AT1)     | Sum of capital instruments meeting the criteria for AT1 and related surplus, additional qualifying minority interest and regulatory adjustments                                           | CET1 + AT1 > 6%          |
| Tier 2 |                             | Sum of capital instruments meeting the criteria for Tier 2 and related surplus, additional qualifying minority interests, qualifying loan loss provisions and regulatory adjustments      | CET1 + AT1 + Tier 2 > 8% |

Basel III reforms have been integrated into the consolidated Basel Framework, which comprises all of the current and forthcoming standards of the Basel Committee on Banking Supervision.

### 7.11.2 Basel III liquidity standards

The introduction of liquidity ratios is one of the main innovations of Basel III. Specifically, the new regulations propose a liquidity coverage ratio (LCR) and a net stable funding ratio (NSFR).

#### *Liquidity coverage ratio (LCR)*

The LCR is designed to ensure that sufficient **high-quality liquid assets** (HQLA) are available for one-month survival in case of a stress scenario. HQLA are defined as cash or assets that can be converted into cash at little or no loss of value in private markets to meet a bank's liquidity needs for a 30 calendar day liquidity stress scenario.

The LCR has two components: (a) the value of the stock of HQLA and (b) the total net outflows, and it is expressed as:

$$LCR = \frac{\text{Stock of HQLA}}{\text{Total net outflows over the next 30 calendar days}} \geq 100\%$$

In order to qualify as HQLA, assets should be liquid in markets during a time of stress and, ideally, be central bank eligible. Assets are considered to be HQLA if they can be easily and immediately converted into cash at little or no loss of value. The liquidity of an asset depends on the underlying stress scenario, the volume to be monetised and the timeframe considered (BCBS, 2019c). According to the BIS definition, HQLA comprise Level 1 and Level 2 assets.<sup>9</sup> Level 1 assets are typically of the highest quality and the most liquid, and generally include cash, central bank reserves and certain marketable securities-backed by sovereigns and central banks, among others. Level 2 assets are comprised of certain government securities,

<sup>9</sup>The EU text was formally published in the Official Journal of the EU on 27 June 2013 (note that the Regulation has also been subject to a subsequent corrigendum). The majority of the rules contained in the legislation are applicable from 1 January 2014.

covered bonds and corporate debt securities. Level 2 assets may also include lower-rated corporate bonds, residential mortgage-backed securities and equities that meet certain conditions (these latter are known as Level 2B assets). Level 2 assets may not in aggregate account for more than 40 per cent of a bank's stock of HQLA. Level 2B assets may not account for more than 15 per cent of a bank's total stock of HQLA.

The total net outflows (over the next 30 days) are defined as:

$$\text{Total net cash outflows} = \text{Total expected cash outflows} - \text{Min (total expected cash inflows; 75 per cent of total expected cash outflows)}$$

The Basel III standard requires that, in normal conditions, the value of the LCR be no lower than 100% (that is, the stock of HQLA should at least equal total net cash outflows). During periods of financial stress, however, banks may use their stock of HQLA, thereby temporarily falling below 100 per cent.

### *The net stable funding ratio (NSFR)*

The NSFR is designed to promote resiliency over longer-term time horizons by creating additional incentives for banks to fund their activities with more stable sources of funding on an ongoing structural basis. In addition, the NSFR aims to limit over-reliance on short-term wholesale funding during times of buoyant market liquidity and encourage better assessment of liquidity risk across all on- and off-balance-sheet items.

The NSFR is expressed as a ratio that must equal or exceed 100 per cent. More specifically, the NSFR is the ratio between the amount of available stable funding (ASF) relative to the amount of required stable funding (RSF):

$$\text{NSFR} = \text{ASF} / \text{RSF}$$

The ASF comprises weighted liabilities reflecting their contractual maturity and is defined as the portion of capital and liabilities expected to be a reliable source of funding over the one-year time horizon considered by the NSFR. The broad characteristics of an institution's funding sources, and their assumed degree of stability are the basis for determining ASF. ASF factors range from 100 per cent – meaning that the funding is expected to be still fully available in more than a year – to 0 per cent – reflecting that funding from this source is unreliable. The RSF of a specific bank is a function of the liquidity characteristics and residual maturities of the various assets held by that institution as well as those of its OBS exposures (BCBS, 2014). These range from 100 per cent to 0 per cent. An RSF factor of 100 per cent means that the asset or exposure needs to be entirely financed by stable funding because it is illiquid. However, an RSF factor of 0 per cent applies to fully liquid and unencumbered assets. The ASF and RSF are calibrated to reflect the presumed degree of stability of a bank's liabilities and liquidity of a bank's assets. The weights for assets and liabilities range from 0 per cent to 100 per cent; these are primarily the result of internationally agreed definitions and calibrations. For example, the NSFR is generally calibrated such that longer-term liabilities are assumed to be more stable than short-term liabilities and that short-term retail deposits are more stable than wholesale funding of the same maturity from other counterparties.

The NSFR became a minimum standard applicable to all internationally active banks on a consolidated basis on 1 January 2018, although national supervisors may also apply it to

<sup>9</sup> The updated definitions of HQLA and LCR can be found in BCBS (2019c) 'LCR liquidity coverage ratio LCR30: High-quality liquid assets', Version effective as of 15 December 2019.

any subset of entities of large internationally active banks or to all other banks (BIS, 2018). Banks must meet the NSFR requirement on an ongoing basis and report on a quarterly basis. Because of its impact on maturity transformation, and since its implementation may have unintended consequences, the NSFR is subject to an observation period which started in 2011. Following this agreement, there has been some discussion – both among policy makers and academics – on whether the two requirements are indeed complementary, and whether both of them are needed to ensure sound liquidity profiles and management (Cecchetti and Kashyap, 2018; Behn *et al.*, 2019).

### 7.11.3 Basel III leverage ratio

In addition to the solvency ratio, Basel III introduced a ‘simple, transparent, non-risk-based’ leverage ratio. This measure intends to restrict the build-up of leverage in the banking sector and reinforce the risk-based requirements with a simple, non-risk-based ‘backstop’ measure (BIS, 2017). The main theoretical justification for the leverage ratio lies in the fact that risk-based ratios cannot completely prevent the undervaluation of certain risks in the denominator. It must be noted that the leverage ratio remains controversial and there remains ambiguity about certain aspects of the exact mechanics. According to the BIS, a leverage ratio is intended to:

- restrict the build-up of leverage in the banking sector to avoid destabilising deleveraging processes that can damage the broader financial system and the economy; and
- reinforce the risk-based requirements with a simple, non-risk-based ‘backstop’ measure.

The Basel Committee is of the view that a simple leverage ratio framework is critical and complementary to the risk-based capital framework and that a credible leverage ratio is one that ensures broad and adequate capture of both the on- and off-balance-sheet leverage of banks. The proposed leverage ratio, expressed as a percentage, is defined as:

$$\text{Leverage ratio} = \frac{\text{Capital measure}}{\text{Exposure measure}}$$

The capital measure is the Tier 1 capital as defined by Basel III. The exposure measure, in addition to on-balance-sheet exposures; will consider derivative exposures, securities financing transaction (SFT) exposures and other off-balance-sheet exposures. The minimum requirement is set at 3 per cent, where it will remain until the BCBS finalises the calibration and makes any necessary adjustments to the definition of the exposure measure. The related public disclosure requirements have been in effect since 1 January 2015.

### 7.11.4 Basel III and G-SIFIs

The Basel III regulations affect all banks; however, the impact may differ across bank type and size. Most banks will be impacted by the increase in quantity and quality of capital, liquidity and leverage ratios, amended Pillar 2 and capital preservation. The more sophisticated banks will be affected by the amended treatment of counterparty credit risk, more robust market risk framework and, to some extent, the amended treatment of securitisations. **Systemically important financial institutions (SIFIs)** and global systemically important financial institutions (**G-SIFIs**) will have to cope with higher capital requirements or be subject to additional



supervision. The terms SIFI, G-SIFI and **G-SIBs** (global systemically important banks) are used to define banks or financial institutions that are deemed too big to fail. The Financial Stability Board has provided a tentative definition as well as a list of the financial institutions considered SIFIs.

G-SIFIs are defined as:

Financial institutions whose distress or disorderly failure, because of their size, complexity and systemic interconnectedness, would cause significant disruption to the wider financial system and economic activity. To avoid this outcome, authorities have all too frequently had no choice but to forestall the failure of such institutions through public solvency support. As underscored by this crisis, this has deleterious consequences for private incentives and for public finances.

(Financial Stability Board, 2011b)

The FSB and BCBS identified an initial group of 29 G-SIBs in 2011 (Financial Stability Board, 2011b). The bucket approach was updated by the Basel Committee document 'Global systemically important banks: Updated assessment methodology and the higher loss absorbency requirement' in 2013. The additional loss-absorbency requirements will initially apply to those banks identified in November 2014 as globally systemically important by the FSB using the BCBS methodology. The methodology used to define the buckets is based on an 'indicator-based' measurement approach. The selected indicators (which include the size of banks, their interconnectedness, lack of readily available substitutes or financial institution infrastructure for the services they provide, their global (cross-border) activity and their complexity) are chosen to reflect the different aspects of what generates negative externalities and makes a bank critical for the stability of the financial system.

In July 2018, the BCBS published a revised version of its assessment methodology, replacing the July 2013 version. The revised assessment methodology will take effect in 2022, and the resulting higher capital buffer requirement will be applied in January 2024.<sup>10</sup> The group of G-SIFIs is updated annually and published by the FSB each November – Table 7.8 illustrates the 2012 and 2020 list. There are a few changes from the original list: compared with the initial group of G-SIBs published, EU banks have either moved to a lower bucket or have been removed from the list. Meanwhile, Chinese banks have either moved to a higher bucket or joined the G-SIB list in more recent years, together with the two largest Canadian banks (Royal Bank of Canada and Toronto Dominion). No bank was allocated to the first two buckets in 2020. The changes in the allocation of the institutions to buckets reflect the effects of changes in underlying activity of banks.

The G-SIBs are subject to more intensive supervision, including stronger supervisory mandates, resources and powers, and higher supervisory expectations for risk management functions, data aggregation capabilities, risk governance and internal controls. Capital requirements for G-SIFIs and G-SIBs will need to have additional loss-absorption capacity tailored to the impact of their default, rising from 1 per cent to 2.5 per cent of risk-weighted assets (with an empty bucket of 3.5 per cent to discourage, in the words of the FSB, further 'systemicness' which is how much the failure of one bank can impact on the rest of the financial system), to be met with common equity.

<sup>10</sup> Details of the indicator-based measurement approach can be found in BCBS (2011c) 'Global systemically important banks: Assessment methodology and the additional loss absorbency requirement'. The revised framework can be found in BCBS (2018b) 'Global systemically important banks: Revised assessment methodology and higher loss absorbency requirement'.

**Table 7.10** G-SIBs and additional capital requirements

| Bucket |       | G-SIBs in alphabetical order within each bucket                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                              |
|--------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |       | 2012                                                                                                                                                                                                                               | 2020                                                                                                                                                                                                                                                                                                                                                         |
| 5      | 3.50% | (empty)                                                                                                                                                                                                                            | (empty)                                                                                                                                                                                                                                                                                                                                                      |
| 4      | 2.50% | Citigroup<br>Deutsche Bank<br>HSBC<br>JPMorgan Chase                                                                                                                                                                               | (empty)                                                                                                                                                                                                                                                                                                                                                      |
| 3      | 2.00% | Barclays<br>BNP Paribas                                                                                                                                                                                                            | Citigroup<br>HSBC<br>JPMorgan Chase                                                                                                                                                                                                                                                                                                                          |
| 2      | 1.50% | Bank of America<br>Bank of New York Mellon<br>Credit Suisse<br>Goldman Sachs<br>Mitsubishi UFJ FG<br>Morgan Stanley<br>Royal Bank of Scotland<br>UBS                                                                               | Bank of America<br>Bank of China<br>Barclays<br>BNP Paribas<br>China Construction Bank<br>Deutsche Bank<br>Industrial and Commercial<br>Bank of China Limited<br>Mitsubishi UFJ FG                                                                                                                                                                           |
| 1      | 1.00% | Bank of China<br>BBVA<br>Groupe BPCE<br>Group Crédit<br>Agricole<br>ING Bank<br>Mizuho FG<br>Nordea<br>Santander<br>Société Générale<br>Standard Chartered<br>State Street<br>Sumitomo Mitsui FG<br>Unicredit Group<br>Wells Fargo | Agricultural Bank of China<br>Bank of New York Mellon<br>Credit Suisse<br>Goldman Sachs<br>Groupe BPCE<br>Group Credit Agricole<br>ING Bank<br>Mizuho FG<br>Morgan Stanley<br>Royal Bank of Canada<br>Santander<br>Société Générale<br>Standard Chartered<br>State Street<br>Sumitomo Mitsui FG<br>Toronto Dominion<br>UBS<br>Unicredit Group<br>Wells Fargo |

FSB member authorities apply the following requirements to G-SIBs:

- *Higher capital buffer*: since the November 2012 update, G-SIBs have been allocated to buckets corresponding to higher capital buffers than they are required to hold by national authorities in accordance with international standards.
- *Total Loss-Absorbing Capacity (TLAC)*: G-SIBs are required to meet the TLAC standard, alongside the regulatory capital requirements set out in the Basel III framework. The TLAC standard has begun being phased-in from 1 January 2019 for G-SIBs identified in the 2015 list that continued to be designated as G-SIBs.
- *Resolvability*: these include group-wide resolution planning and regular resolvability assessments. The resolvability of each G-SIB is also reviewed in a high-level FSB

resolvability assessment process (RAP) by senior regulators within the firms' crisis management groups.

- *Higher supervisory expectations*: these include supervisory expectations for risk management functions, risk data aggregation capabilities, risk governance and internal controls.

## 7.12 Finalising Basel III

In December 2017, the BCBS published the final instalments of its reforms for the calculation of risk-weighted assets (RWA) and capital floors. While Basel III focused on the reform of regulatory capital, Basel IV changes the approaches for the calculation of RWA, regardless of risk type and irrespective of whether standardised approaches or internal models are used. The 2017 reforms aim to restore credibility in the calculation of RWA and improve the comparability of banks' capital ratios (BIS, 2017). According to the BIS, the reforms were necessary to address any remaining weaknesses in the regulatory framework and increase its credibility. They should also allow for the use of internal models while minimising regulatory arbitrage. The reforms introduce constraints on the estimates banks make when they use their internal models for regulatory capital purposes and, in some cases, remove the use of internal models. The 2017 reforms also introduce a streamlined treatment of operational risk, replacing the existing four approaches with a single standardised approach. Finally, the 2017 reforms introduce a leverage ratio surcharge for the largest banks (G-SIBs), set at 50 per cent of their risk-based capital buffer. For example, a bank with a 2 per cent risk-based buffer will have a 1 per cent leverage ratio buffer and therefore it will be expected to maintain a leverage ratio of at least 4 per cent. The BIS argues that the leverage ratio buffer is necessary to make sure that the leverage ratio continues to act as an appropriate backstop to the risk-based requirements for G-SIBs. The implementation of the next-generation of reforms creates a remarkable challenge for the banking sector, but banks will have some time to

**Table 7.11** Finalising Basel III: a roadmap

| 2017 reforms                                             | Implementation date                                                                                                                                                 |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Revised standardised approach for credit risk            | 1 January 2022                                                                                                                                                      |
| Revised internal ratings-based framework for credit risk | 1 January 2022                                                                                                                                                      |
| Revised credit-valuation adjustment framework            | 1 January 2022                                                                                                                                                      |
| Revised operational risk framework                       | 1 January 2022                                                                                                                                                      |
| Revised market risk framework                            | 1 January 2022                                                                                                                                                      |
| Leverage ratio                                           | Revised exposure definition: 1 January 2022<br>G-SIBs buffer: 1 January 2022                                                                                        |
| Output floor                                             | 1 January 2022: 50%<br>1 January 2023: 55%<br>1 January 2024: 60%<br>1 January 2025: 65%<br>1 January 2026: 70%<br>1 January 2027: 72.5% (steady-state calibration) |

prepare. Table 7.11 summarises the key changes introduced by the 2017 reforms, as well as their implementation timetable. It is worth noting that, at the time of writing (May 2021), the Covid-19 pandemic might cause delays and postponements.

To summarise, the 2017 reforms aim to enhance risk capture by revising the standardised approaches for calculating credit and market risk, credit valuation adjustment and operation risk. The aim is to achieve greater risk sensitivity and greater comparability across banks. Constraints on using internal models aim to reduce unwarranted variability in banks' calculations of RWAs. Output floors aim to limit the benefits that banks can derive from using internal models to calculate minimum capital requirements. Finally, the 2017 reforms aim to constrain bank leverage, by introducing a leverage surcharge for G-SIBs.

## 7.13 Conclusion

In this chapter we have reviewed the issue of financial regulation. The chapter began with a review of the rationale for regulation, introducing the reader to different types of regulation. The limitations of regulation were also analysed, in particular the moral hazard issue connected with the government safety net arrangements such as deposit insurance and the lender of last resort function. The chapter also focused on the Basel Capital Accord and the effort of the Basel Committee to provide common regulatory standards for internationally active banks. Bank regulation cannot prevent financial crises, but the regulatory framework that is currently being shaped will influence the development of the banking system for many years to come. Measures by governments to purchase impaired assets, recapitalise troubled banks and inject liquidity into the system have commanded support among many banking academics and practitioners.

### Key terms

|                                               |                                   |                        |                                       |
|-----------------------------------------------|-----------------------------------|------------------------|---------------------------------------|
| Agency capture                                | Deposit insurance                 | Lender of last resort  | SIFIs                                 |
| Bank runs                                     | Financial                         | Macro-prudential       | Supervision                           |
| Bank Recovery and Resolution Directive (BRRD) | safety net                        | regulation             | Systemic risk                         |
| Capital adequacy                              | G-SIBs                            | Micro-prudential       | Systemic stability                    |
| Compliance costs                              | G-SIFIs                           | regulation             | Too big to fail/Too important to fail |
| Consumer protection                           | Government                        | Monitoring             |                                       |
| Contagion                                     | safety net                        | Regulation             |                                       |
|                                               | High-quality liquid assets (HQLA) | Regulatory forbearance |                                       |

## Key reading

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## REVISION QUESTIONS AND PROBLEMS

- 7.1** Is there a rationale for the regulation of financial intermediaries and financial markets?
- 7.2** What is a bank run?
- 7.3** What are the main types of financial regulation?
- 7.4** Why are the 'safety net' arrangements said to increase moral hazard in financial markets?
- 7.5** What are the main limitations of financial regulation?
- 7.6** What is regulatory forbearance? Describe the main costs and benefits of engaging in forbearance.
- 7.7** What is the financial safety net? Why is deposit insurance the central element of a well functioning financial safety net?
- 7.8** What are the main drivers of regulatory reforms?
- 7.9** Discuss the 'too big to fail' hypothesis. Can you give some recent examples?
- 7.10** Illustrate the main features of the Basel Capital Accords, with a focus on the reforms introduced by Basel III.